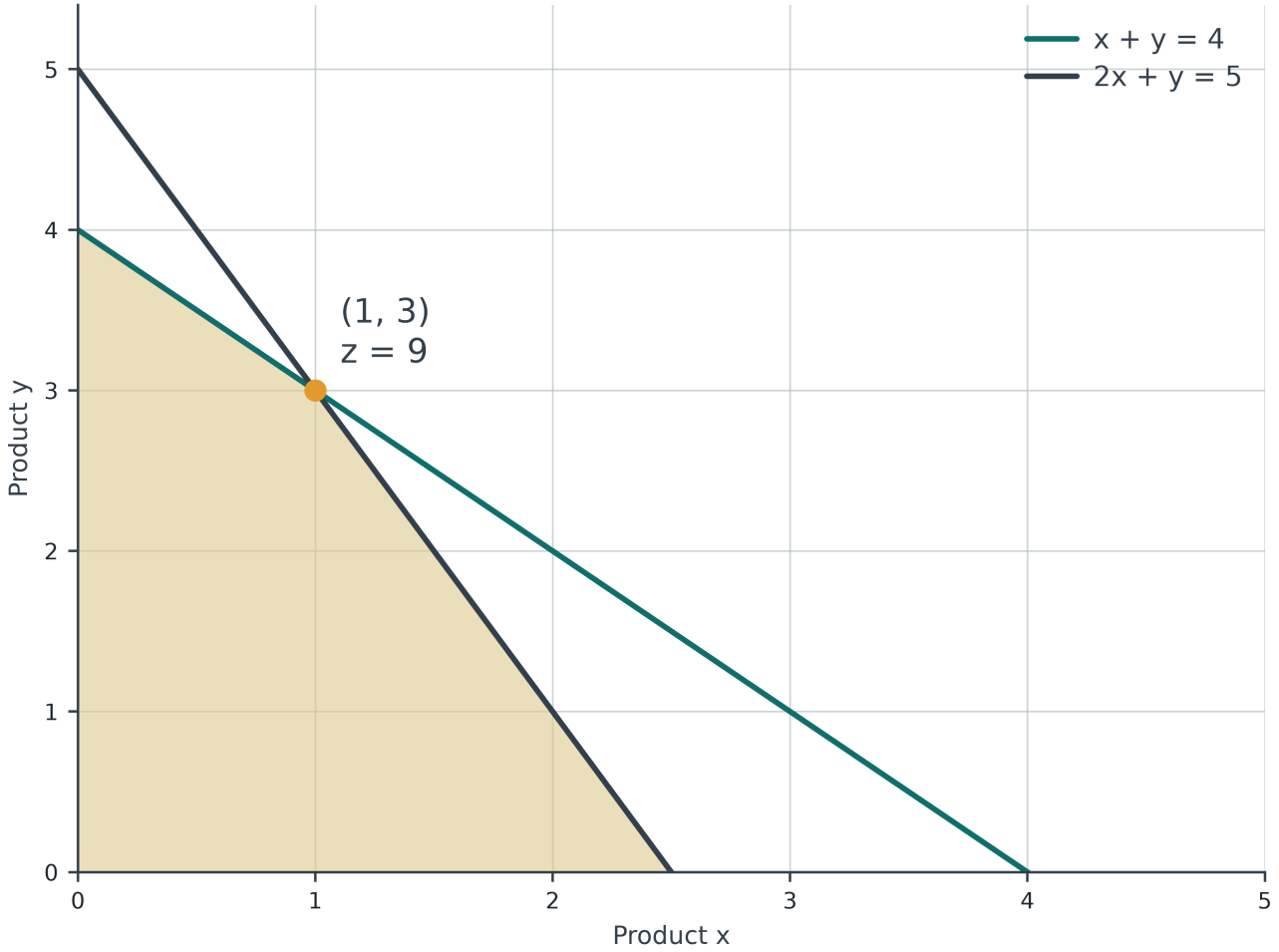
Linear programming and duality

2105623 Optimization of Chemical Processes

26 October 2026 · 180-minute class

Geometry, Simplex, optimality certificates, and resource values



Plan for the three-hour class

Minutes	Topic
0–27	LP geometry and corner solutions
27–62	Basic feasible solutions and Simplex
62–72	Break
72–113	Duality and complementary slackness
113–166	Shadow prices and sensitivity
166–180	Exit ticket, solutions, and recap

What makes a model linear?

$\max c^T x$ subject to $Ax \le b, x \ge 0$

Coefficients are fixed; decision variables are continuous.

$x + 2y$ is linear. xy and x^2 are not.

A binary restriction changes an LP into a MILP.

Running example: two production routes

Item	Route x	Route y	Daily limit
Resource 1	1 unit/t	1 unit/t	4 units/day
Resource 2	2 units/t	1 unit/t	5 units/day
Contribution	3 × $100/t	2 × $100/t	Maximize

x, y = tonnes/day; contribution = $100/day units.

Choose continuous production; no fixed startup cost.

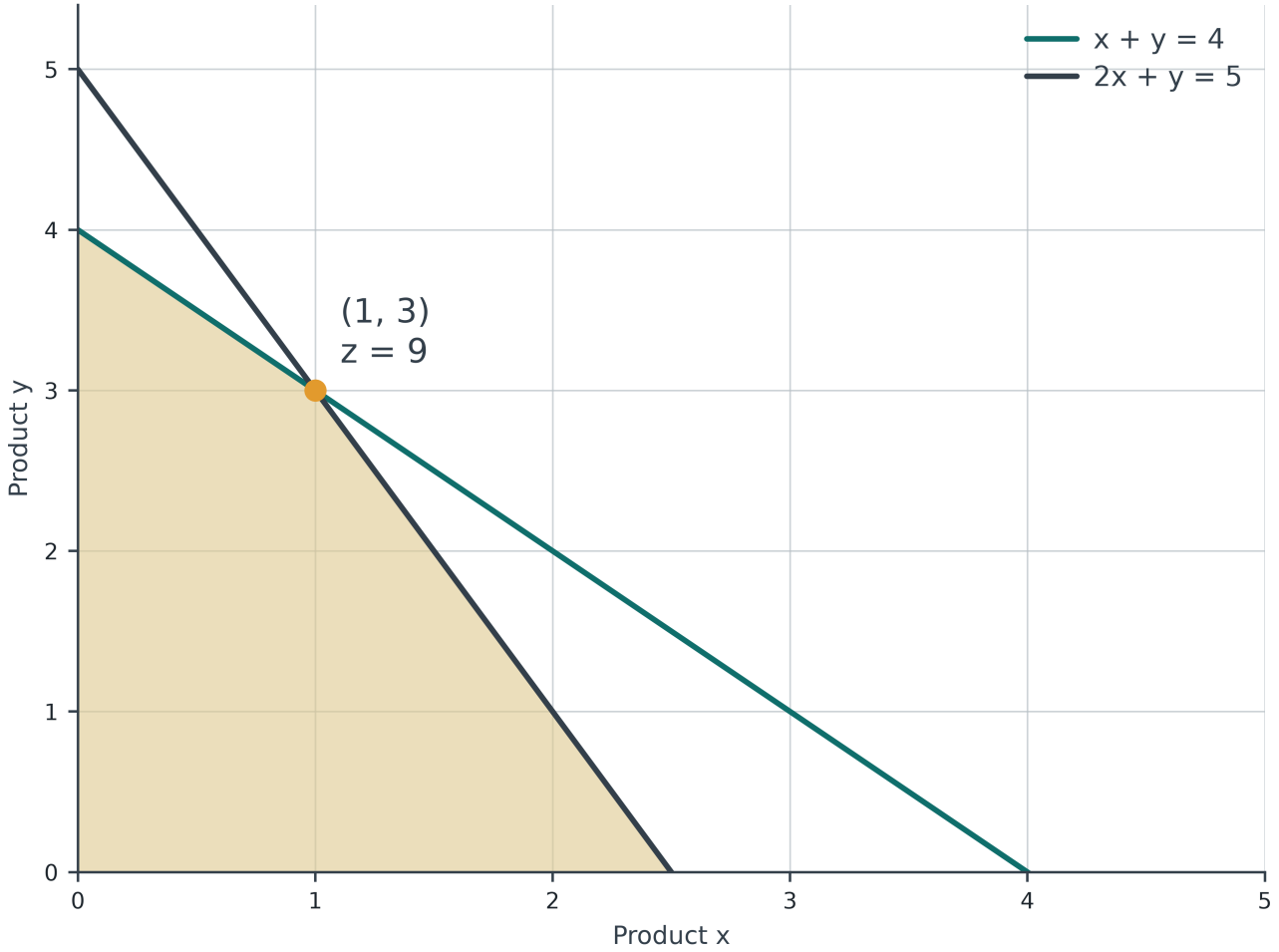
The production LP

$\max z = 3x + 2y$

$x + y \leq 4$

$2x + y \leq 5$

$x \geq 0, y \geq 0$



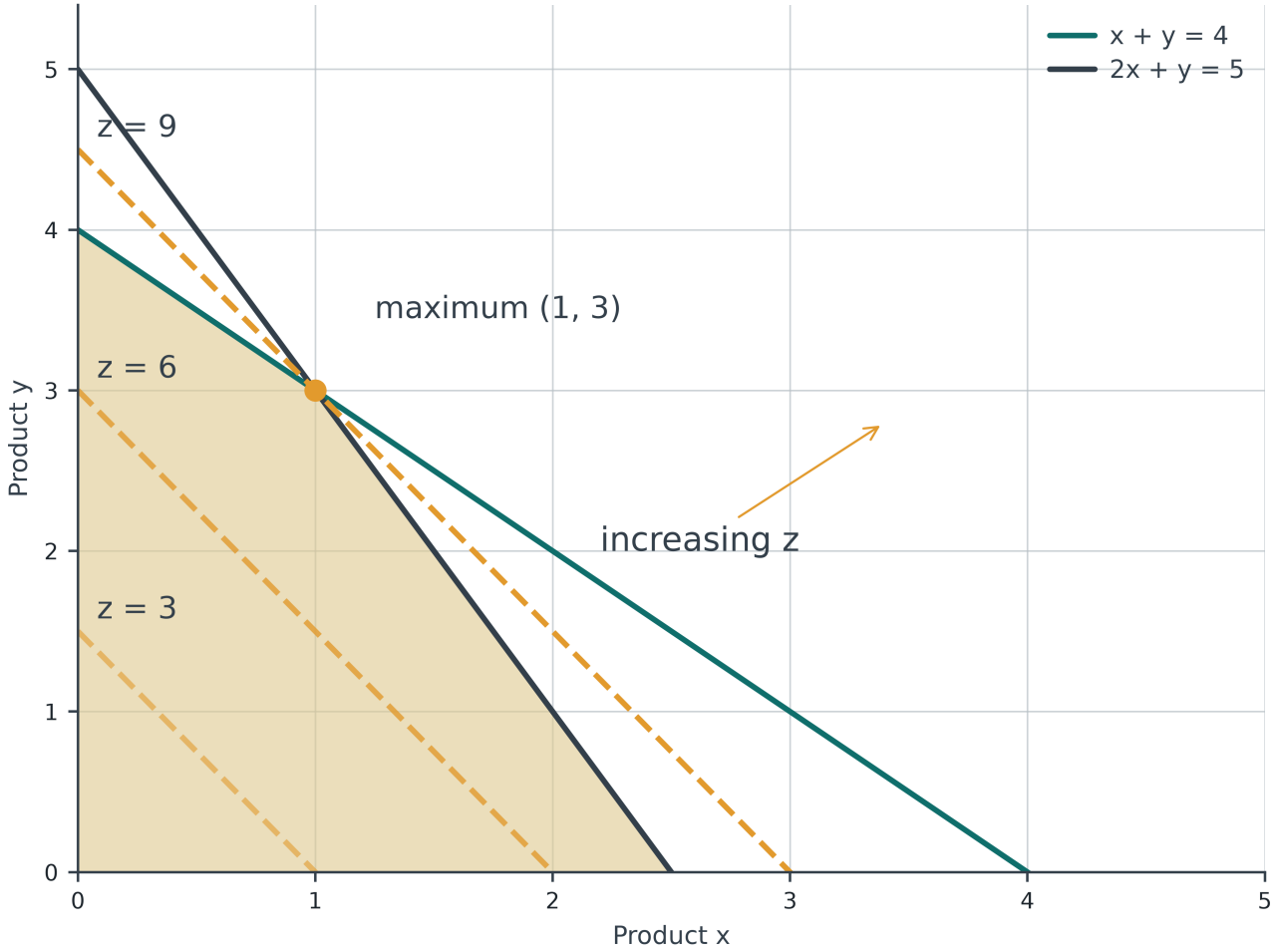
Objective contours and the best feasible point

$3x + 2y = z$ gives parallel lines.

Move the line toward larger z.

The last contact is $(x^*, y^*) = (1, 3)$.

$z^* = 9$, or \$900/day.



Why compare extreme points?

Vertex (x, y)	z = 3x + 2y
(0, 0)	0
(2.5, 0)	7.5
(1, 3)	9
(0, 4)	8

On a nonempty bounded polytope, a linear objective has an optimal extreme point.

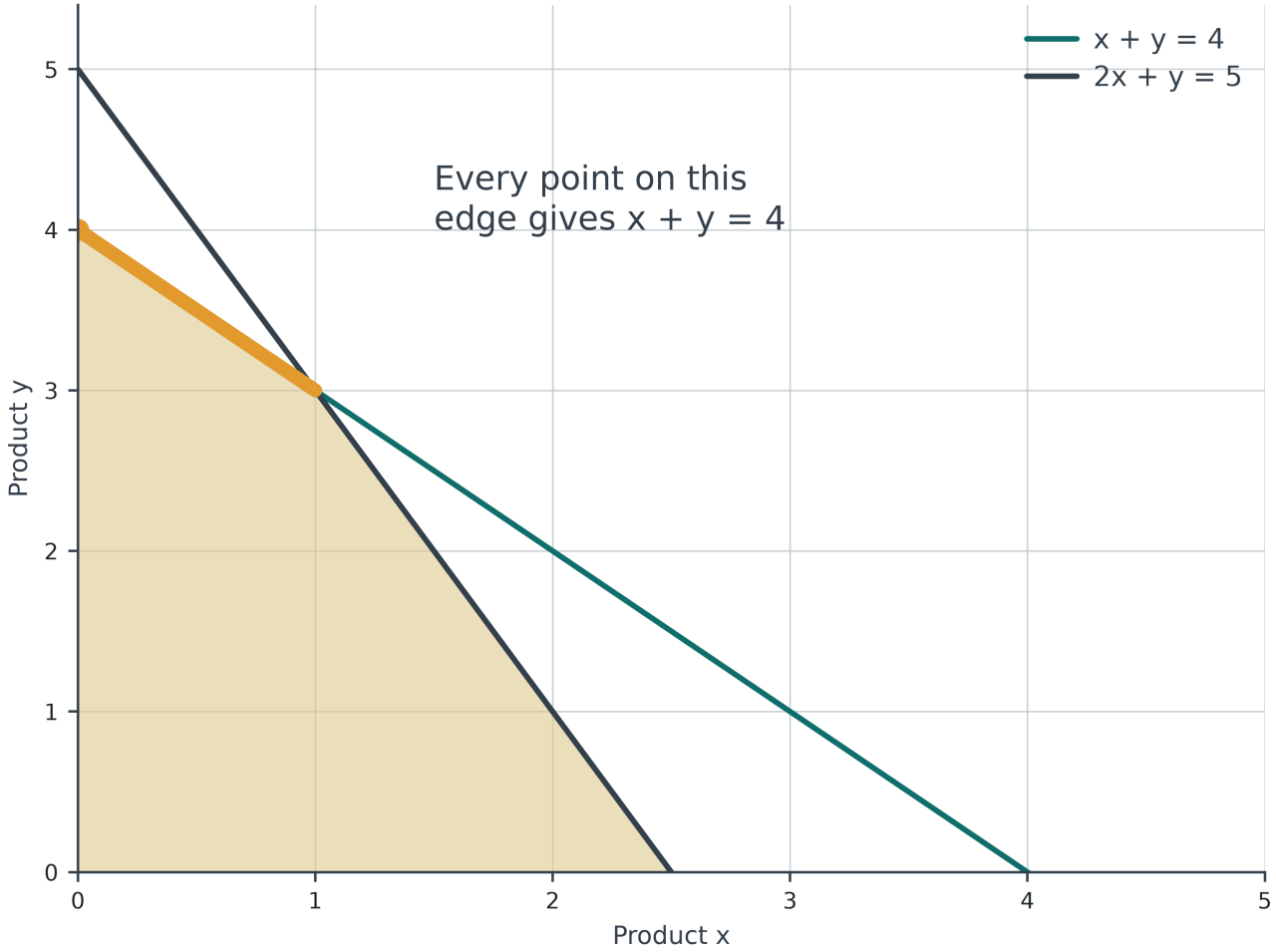
A convex combination cannot beat all vertices used in that combination.

An LP can have many optimal solutions

Change only the objective to max x + y.

Every point from (0,4) to (1,3) is optimal.

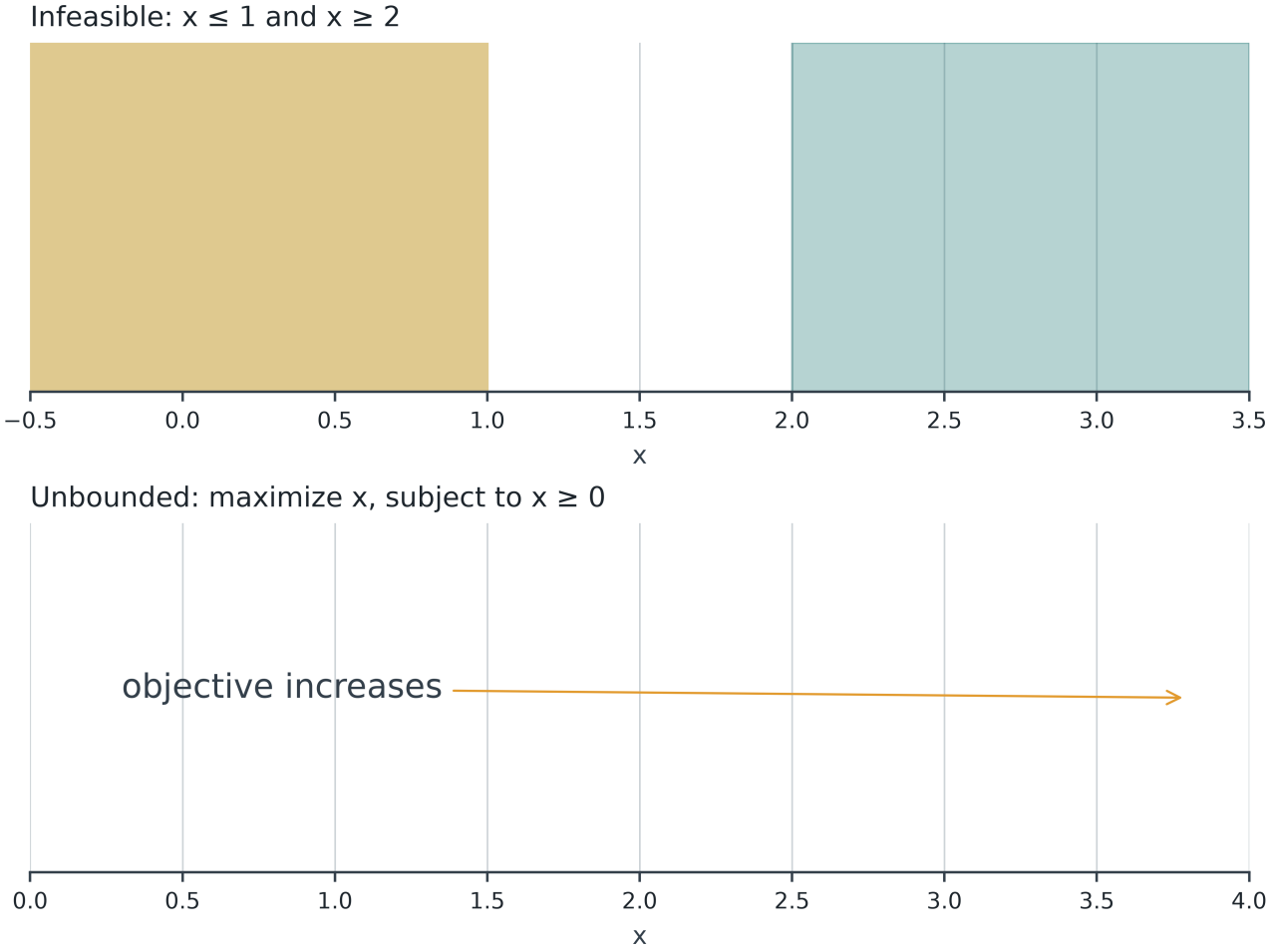
The entire optimal edge has value 4.



Infeasible and unbounded are different

Infeasible: $x \le 1$ and $x \ge 2$ cannot both hold.

Unbounded: max x with only $x \ge 0$ has no finite maximum.



Quick check: feasibility before profit

Original LP: max $3x+2y$; $x+y\le4$, $2x+y\le5$, $x,y\ge0$.

A: (2,2) gives $z=10$. B: (1,3) gives $z=9$.

Which point can be accepted? Explain the failed constraint.

Slack variables and standard form

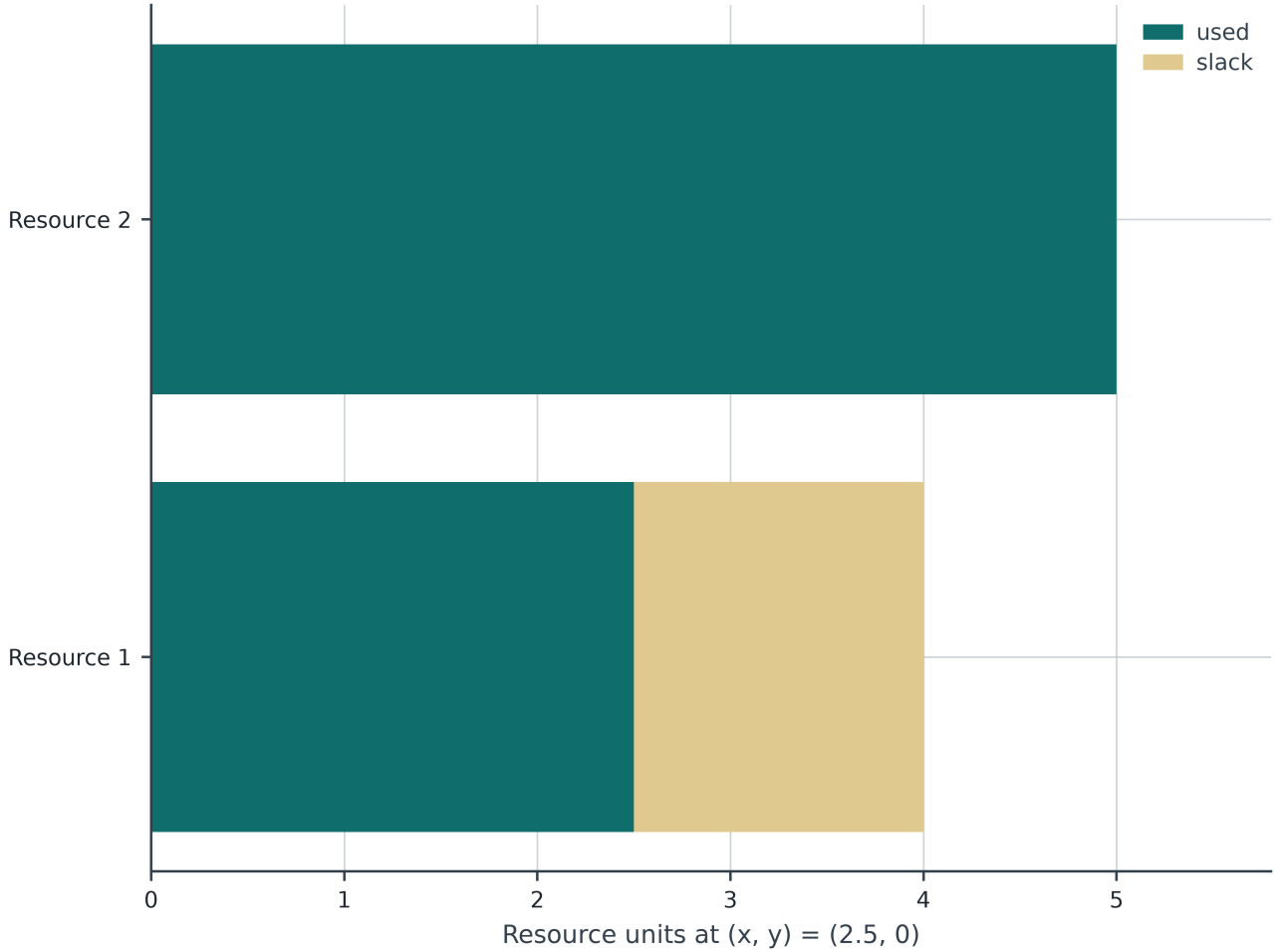
$x + y + s_1 = 4$

$2x + y + s_2 = 5$

$x, y, s_1, s_2 \geq 0$

s_1 and s_2 measure unused resources.

Shown: $(x,y)=(2.5,0)$, so $(s_1,s_2)=(1.5,0)$.



A basis selects the variables we solve for

Basis	Nonbasic variables	Basic solution
$\{s_1, s_2\}$	$x=y=0$	$s_1=4,\ s_2=5$
$\{x, s_1\}$	$y=s_2=0$	$x=2.5,\ s_1=1.5$
$\{x, y\}$	$s_1=s_2=0$	$x=1,\ y=3$

Choose two independent columns as a basis for the two equations.

Set the other two variables to zero, then solve.

A basic solution is feasible only if every variable is nonnegative.

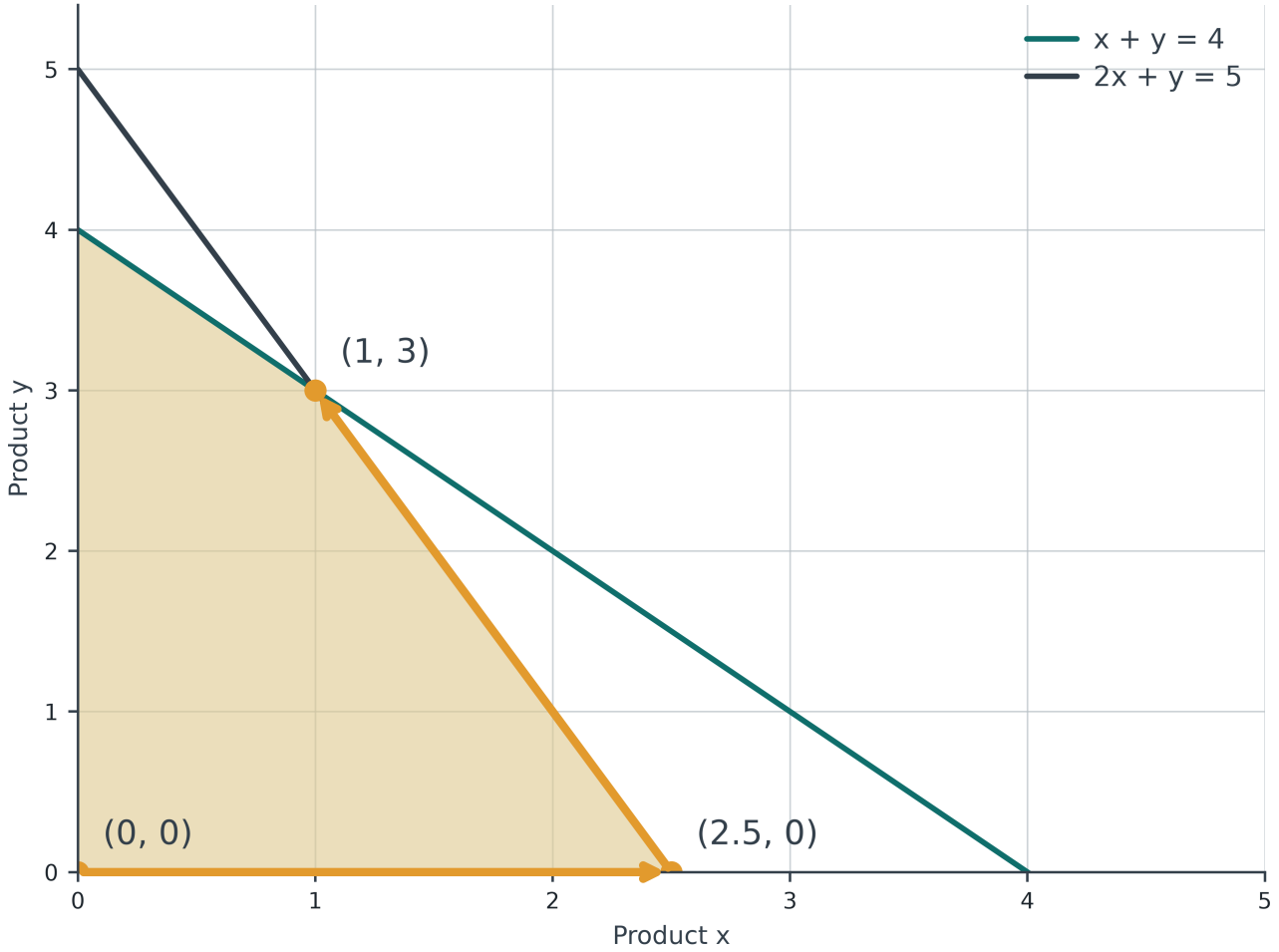
The Simplex decision at each step

1. Choose an improving nonbasic variable.

2. Increase it until a basic variable reaches zero.

3. Pivot: exchange the entering and leaving variables.

Positive reduced cost improves z if a positive feasible step exists.



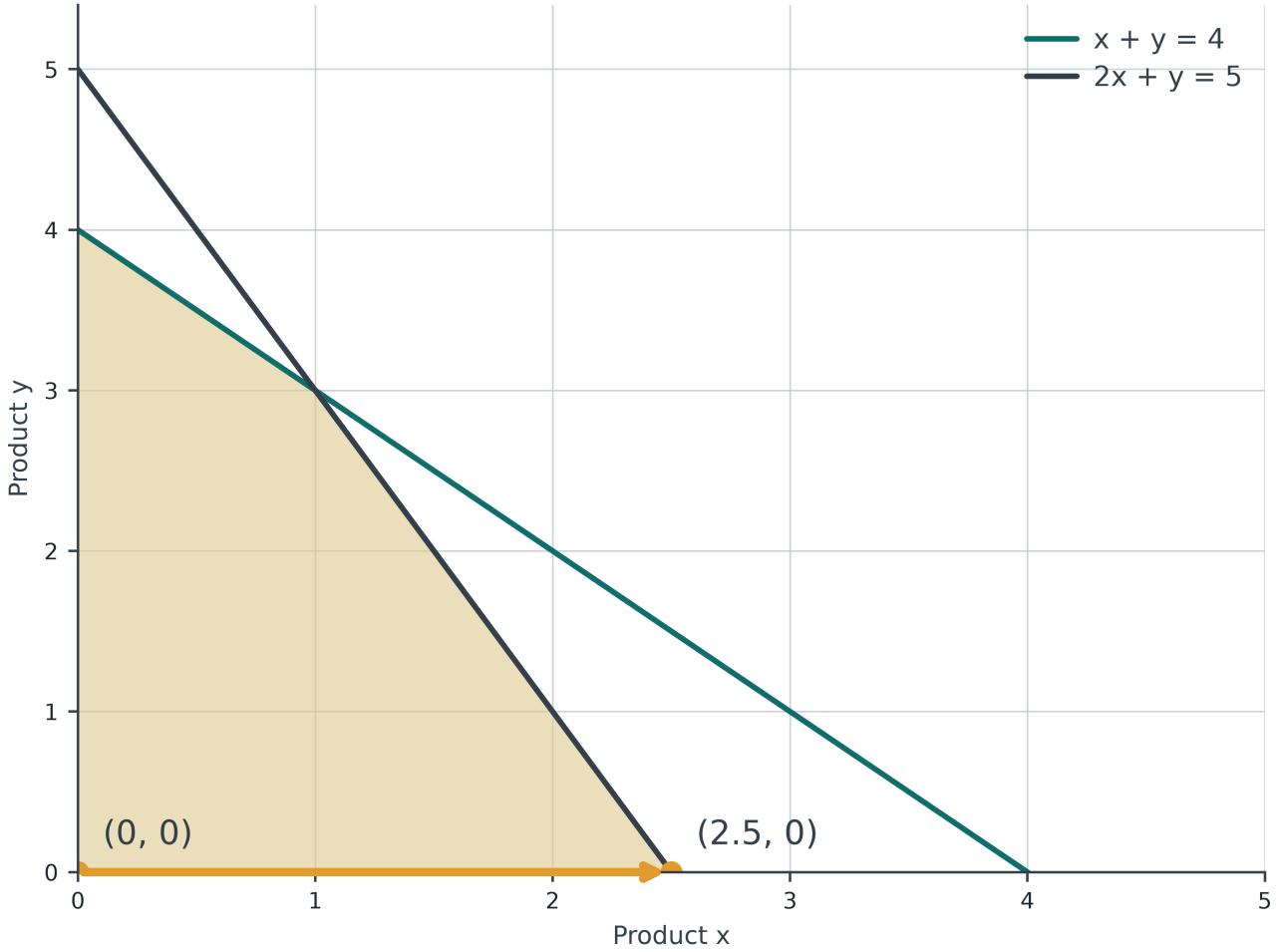
Pivot 1: x enters, s_2 leaves

At the origin: $z = 3x + 2y$. Choose x.

With $y=0$: $s_1=4-x$; $s_2=5-2x$.

Ratio test: $4/1=4$; $5/2=2.5$.

Move to (2.5,0); $z=7.5$.



Dictionary after the first pivot

$x = 2.5 - 0.5y - 0.5s_2$

$s_1 = 1.5 - 0.5y + 0.5s_2$

$z = 7.5 + 0.5y - 1.5s_2$

Nonbasic: $y = s_2 = 0$. The $+0.5$ coefficient means y can improve z.

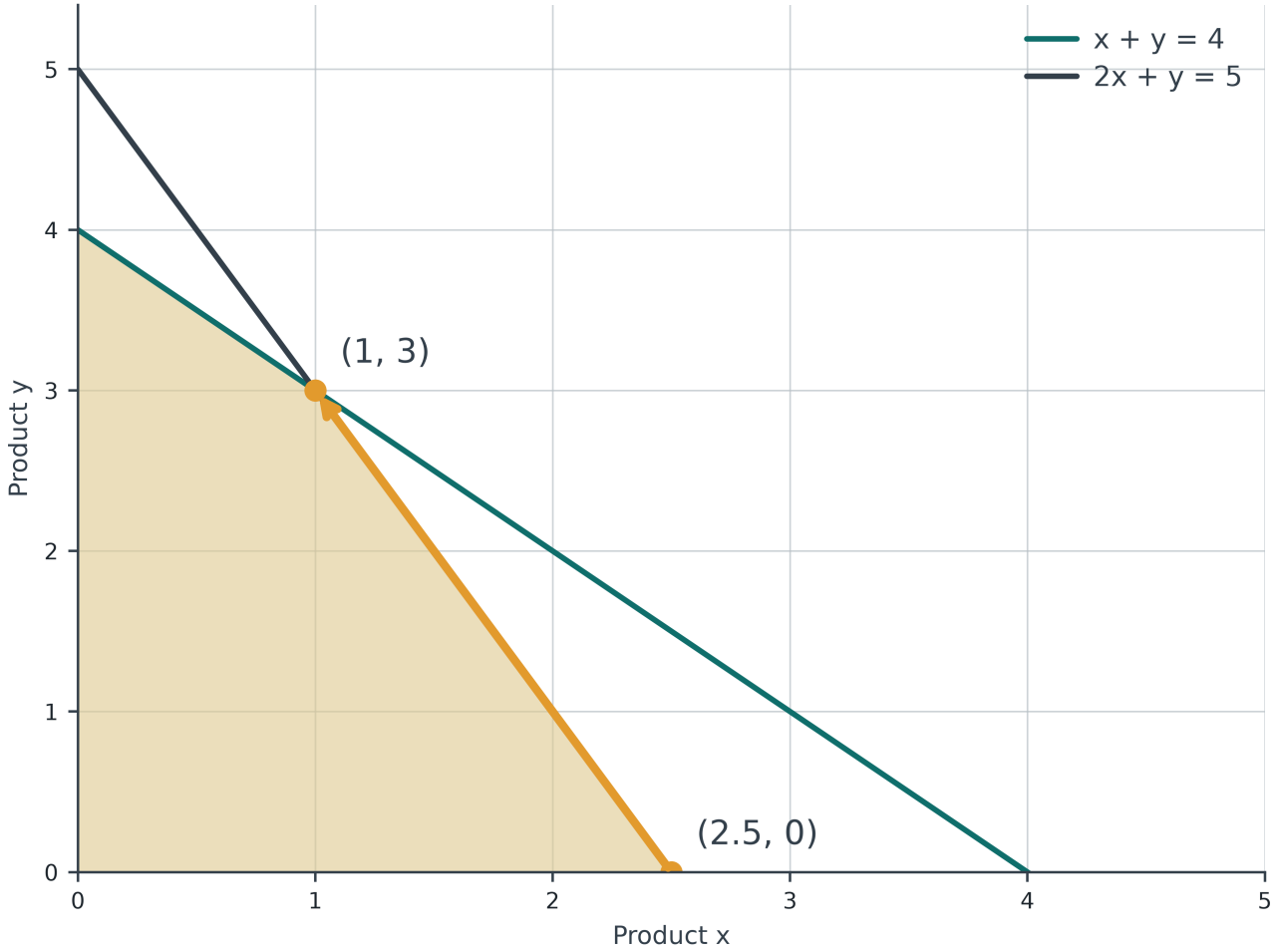
Pivot 2: y enters, s_1 leaves

Hold $s_2=0$ and increase y.

$x \geq 0$ gives $y \leq 2.5/0.5=5$.

$s_1 \geq 0$ gives $y \leq 1.5/0.5=3$.

Thus $y=3$, $x=1$, $s_1=s_2=0$; $z=9$.



The final dictionary certifies optimality

$$x = 1 + s_1 - s_2$$

$$y = 3 - 2s_1 + s_2$$

$$z = 9 - s_1 - s_2$$

Because $s_1, s_2 \geq 0$, every feasible point satisfies $z \leq 9$.

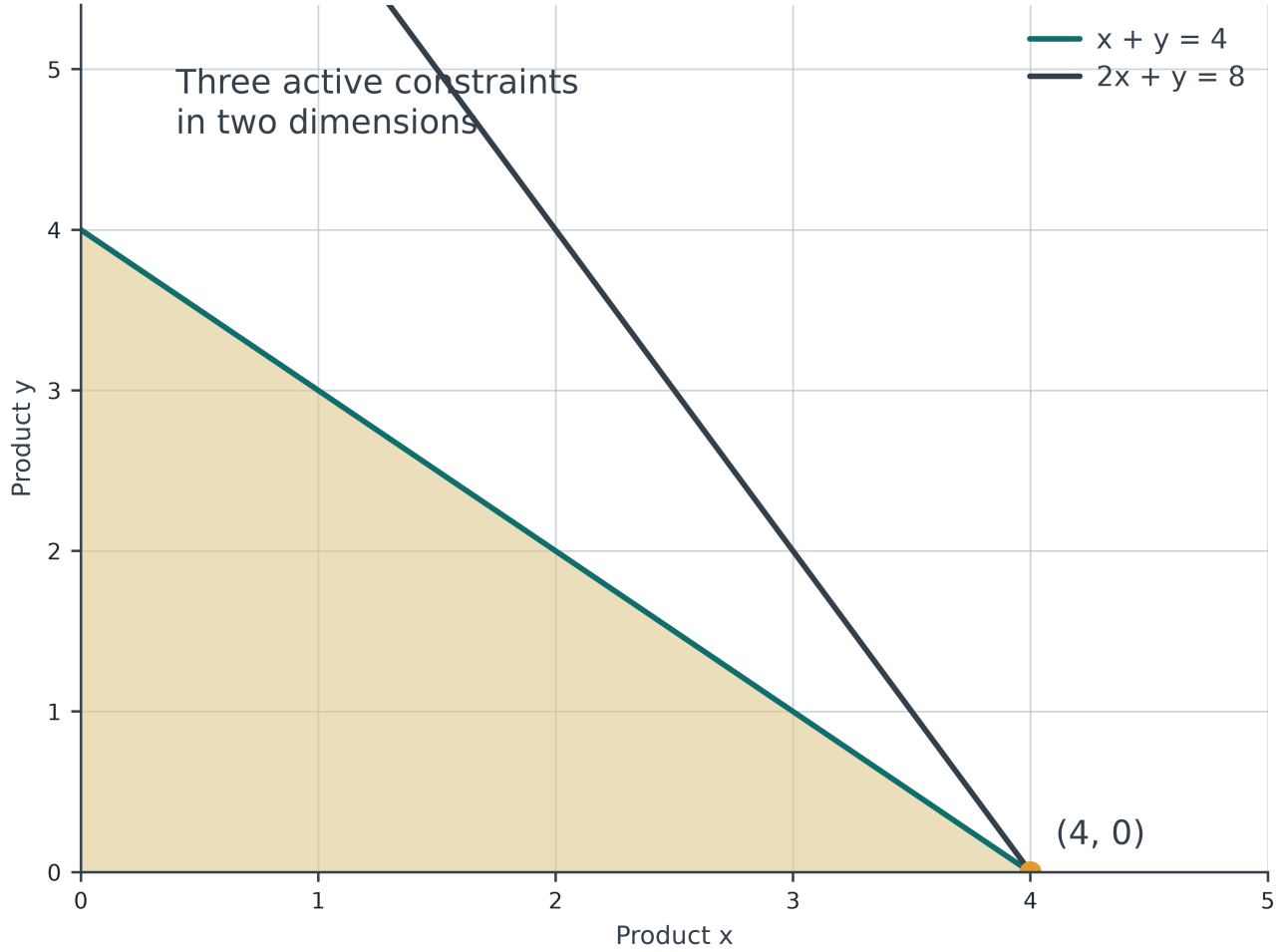
Degeneracy and starting feasibility

A degenerate BFS has at least one zero basic variable.

A pivot may change the basis without improving the point.

At b=(4,8), the basis {x,y} gives x=4, y=0.

If a feasible slack basis is unavailable, use Phase I or another initialization.



Practice: change the contribution

Keep $x+y\le 4$, $2x+y\le 5$, $x,y\ge 0$.

Change the objective to $\max z=x+3y$.

Choose the first entering variable and apply the ratio test.

Predict the optimum and write a stopping argument.

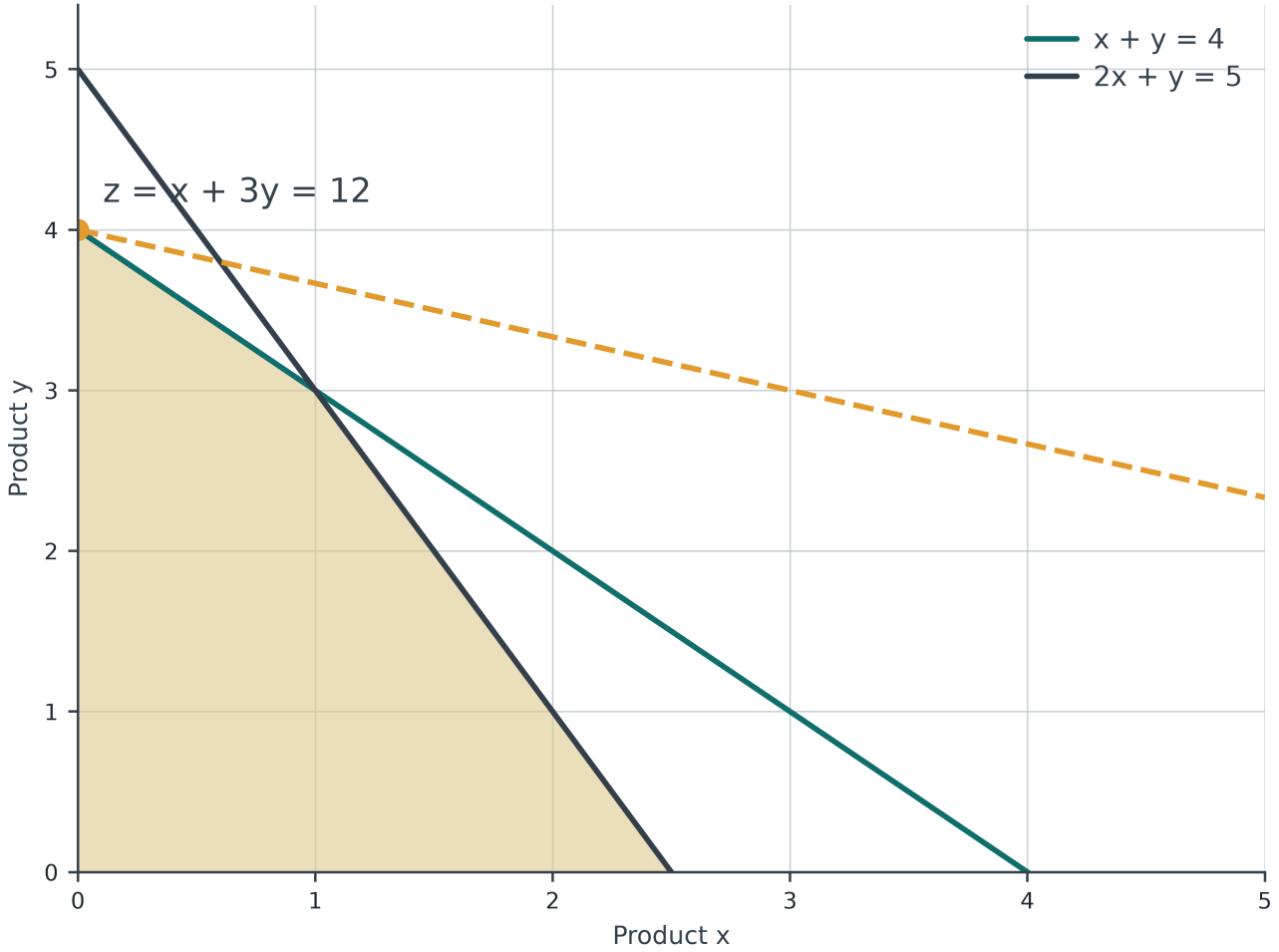
Practice solution: one pivot is enough

y enters; min(4/1,5/1)=4, so s_1 leaves.

$y=4-x-s_1$; $s_2=1-x+s_1$.

$z=x+3y=12-2x-3s_1\leq 12$.

Optimal: (x,y)=(0,4), slack=(0,1), $z^*=12$.



Break

10 minutes

After the break: what is one more unit of a resource worth?

The dual prices the resources

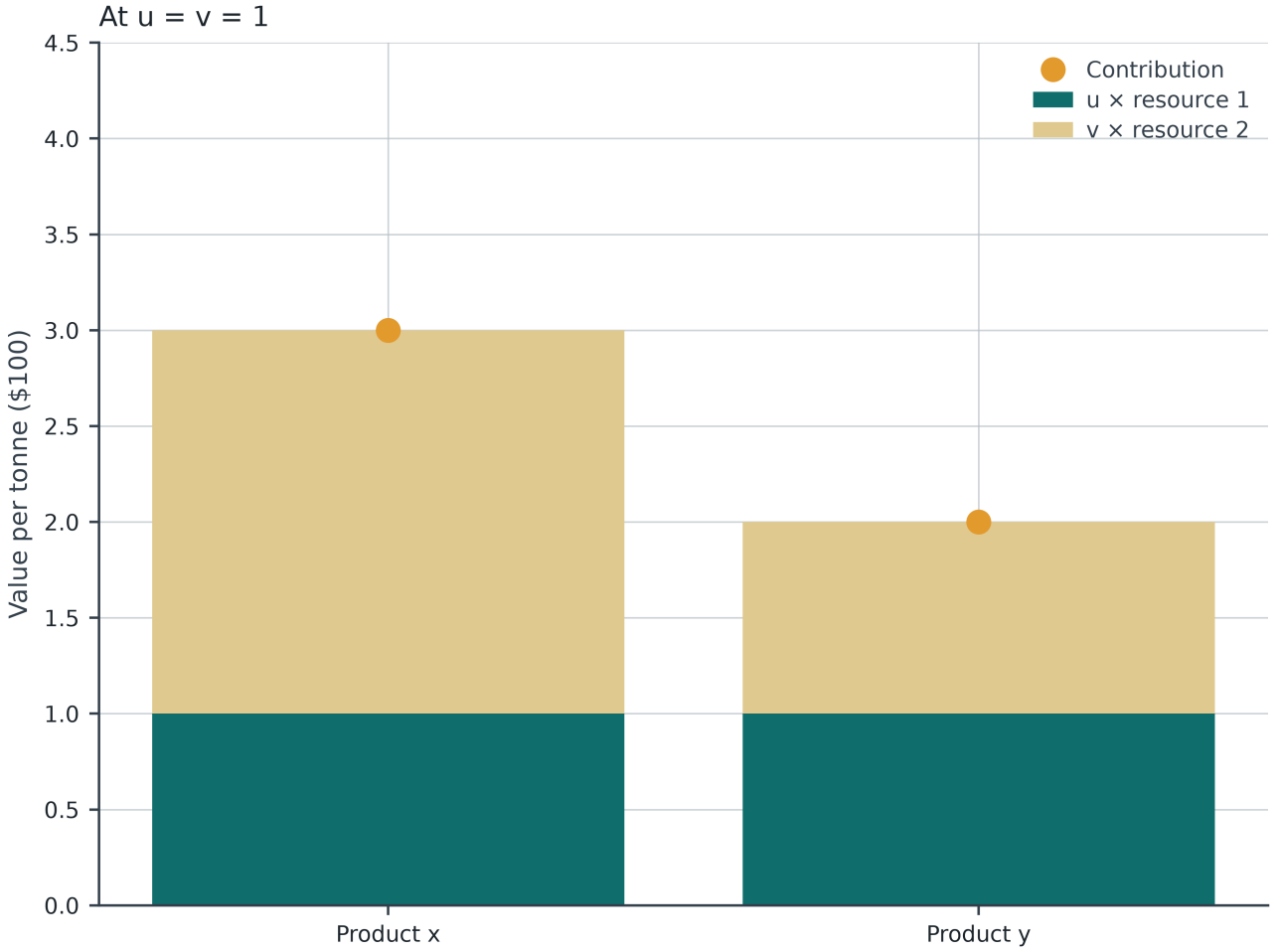
Return to max z=3x+2y, b=(4,5).

u, v price resources 1 and 2 ($100/unit).

A tonne of x uses resources worth u+2v.

A tonne of y uses resources worth u+v.

Cover contributions: u+2v≥3, u+v≥2.



Primal and dual side by side

Primal: max $c^T x$, subject to $Ax \leq b$, $x \geq 0$.

Dual: min $b^T u$, subject to $A^T u \geq c$, $u \geq 0$.

Here: min $w = 4u + 5v$

subject to $u + 2v \geq 3$, $u + v \geq 2$, $u, v \geq 0$.

Signs depend on the formulation

For a maximization primal:

Primal feature	Dual counterpart
Constraint $a_i^T x \le b_i$	$u_i \ge 0$
Constraint $a_i^T x \ge b_i$	$u_i \le 0$
Constraint $a_i^T x = b_i$	u_i is free
Variable $x_j \ge 0$	$(A^T u)_j \ge c_j$
Variable $x_j \le 0$	$(A^T u)_j \le c_j$
Variable x_j is free	$(A^T u)_j = c_j$

Weak duality gives an upper bound

For any primal-feasible x and dual-feasible u:

$c^T x \le x^T A^T u = u^T A x \le u^T b$.

Example: (u,v)=(2,1) is feasible; w=13.

Therefore every feasible production plan has z≤13.

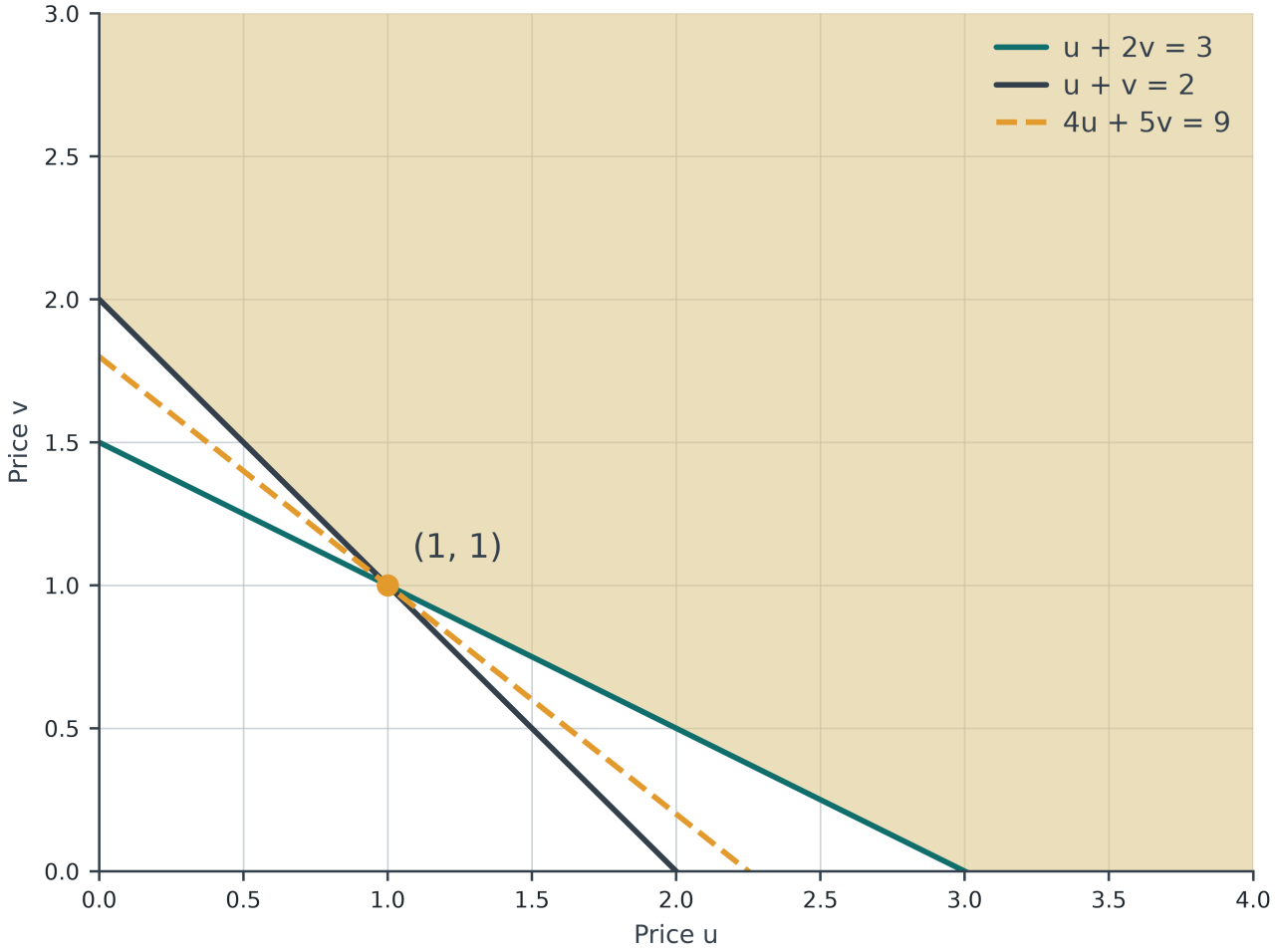
Dual geometry: tighten the upper bound

The dual feasible set lies above both price lines.

Minimize 4u+5v over that set.

Candidate corner prices: (3,0), (1,1), (0,2).

Their bounds are 12, 9, and 10.



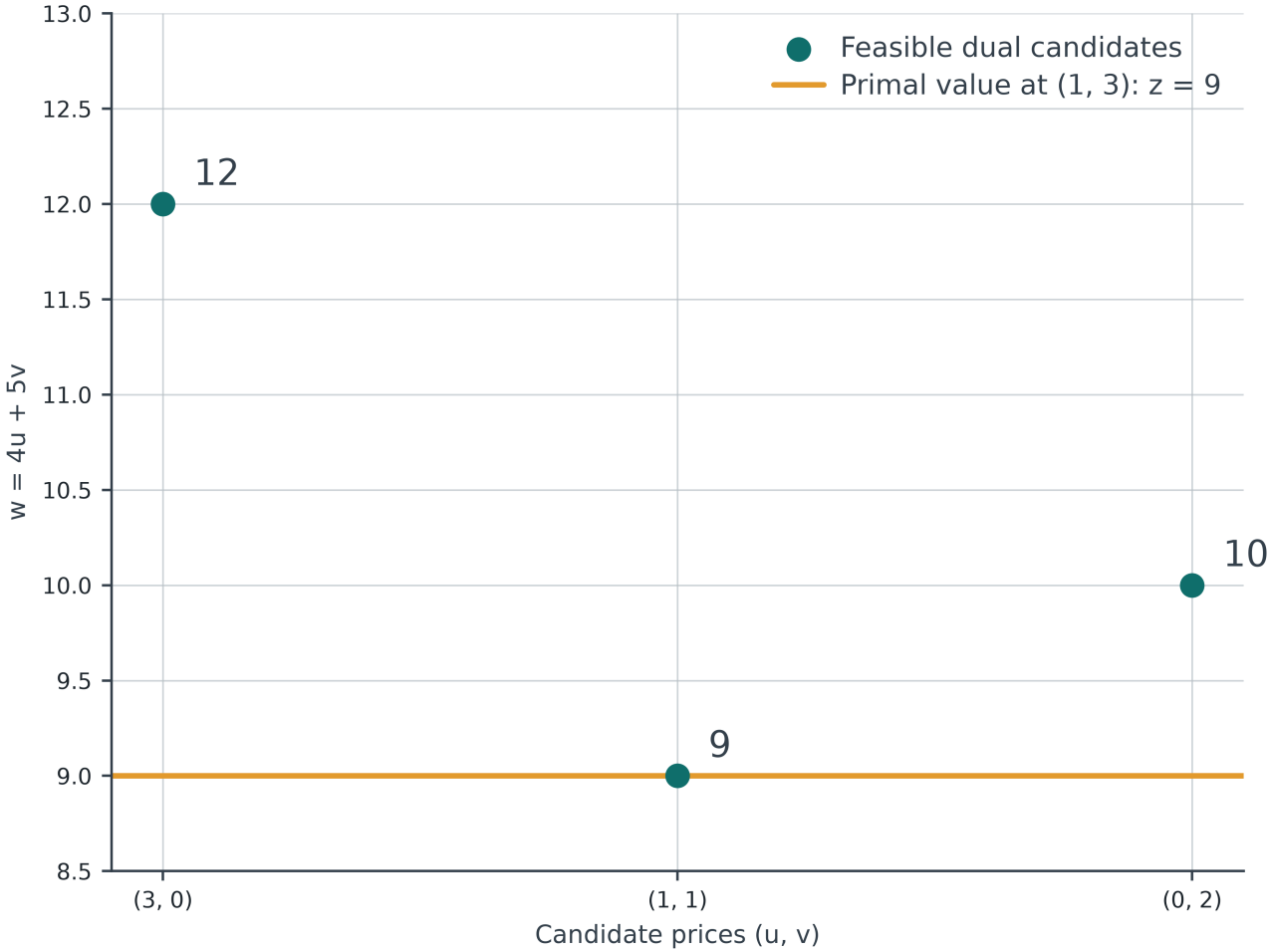
Solve the dual and compare

$u+2v=3$ and $u+v=2 \rightarrow u=v=1$.

Dual value: $4(1)+5(1)=9$.

Primal point (1,3) is feasible and gives 9.

Feasible values meet: both solutions are optimal.



Strong duality for linear programs

If an LP is feasible with a finite optimum, its dual also has an optimum with the same value.

A primal unbounded above implies the dual is infeasible.

Primal infeasible does not by itself imply dual unbounded.

The duality gap separates two kinds of waste

$s = b - Ax \geq 0$; $r = A^T u - c \geq 0$.

$w - z = u^T s + x^T r \geq 0$.

Unused priced resources contribute $u^T s$.

Producing below the assigned resource cost contributes $x^T r$.

Complementary slackness

For primal-feasible x and dual-feasible u:

$u_i(b_i - a_i^T x) = 0$ for each resource.

$x_j((A^T u)_j - c_j) = 0$ for each product.

Unused resource implies zero price. Positive output implies a tight dual constraint.

A binding resource can have zero price.

LP duality is KKT in a special form

Rewrite $\max c^T x$ as $\min -c^T x$.

$L = -c^T x + u^T(Ax - b) - r^T x$.

Stationarity: $-c + A^T u - r = 0$.

Primal feasibility, $u, r \geq 0$, and complementarity complete the optimality conditions.

Practice: a dual certificate

For max x+3y with the same two resource constraints:

1. Write the dual.

2. Test primal (0,4) and dual (u,v)=(3,0).

3. Check both objectives and all complementarity products.

Practice solution: zero price for unused capacity

Dual: min $4u+5v$; $u+2v\geq 1$, $u+v\geq 3$; $u,v\geq 0$.

At $(3,0)$: $w=12=z(0,4)$.

Resource slack: $(0,1)$. Dual slack: $(2,0)$.

Products: $3\cdot 0=0$, $0\cdot 1=0$, $0\cdot 2=0$, $4\cdot 0=0$.

Shadow prices: value per extra unit

Return to c=(3,2), b=(4,5): u*=v*=1.

Hold A and c fixed; keep the same basis feasible and optimal.

Then $\Delta z^* = u^*\Delta b_1 + v^*\Delta b_2$ exactly.

Each additional resource unit is worth $100 within this region.

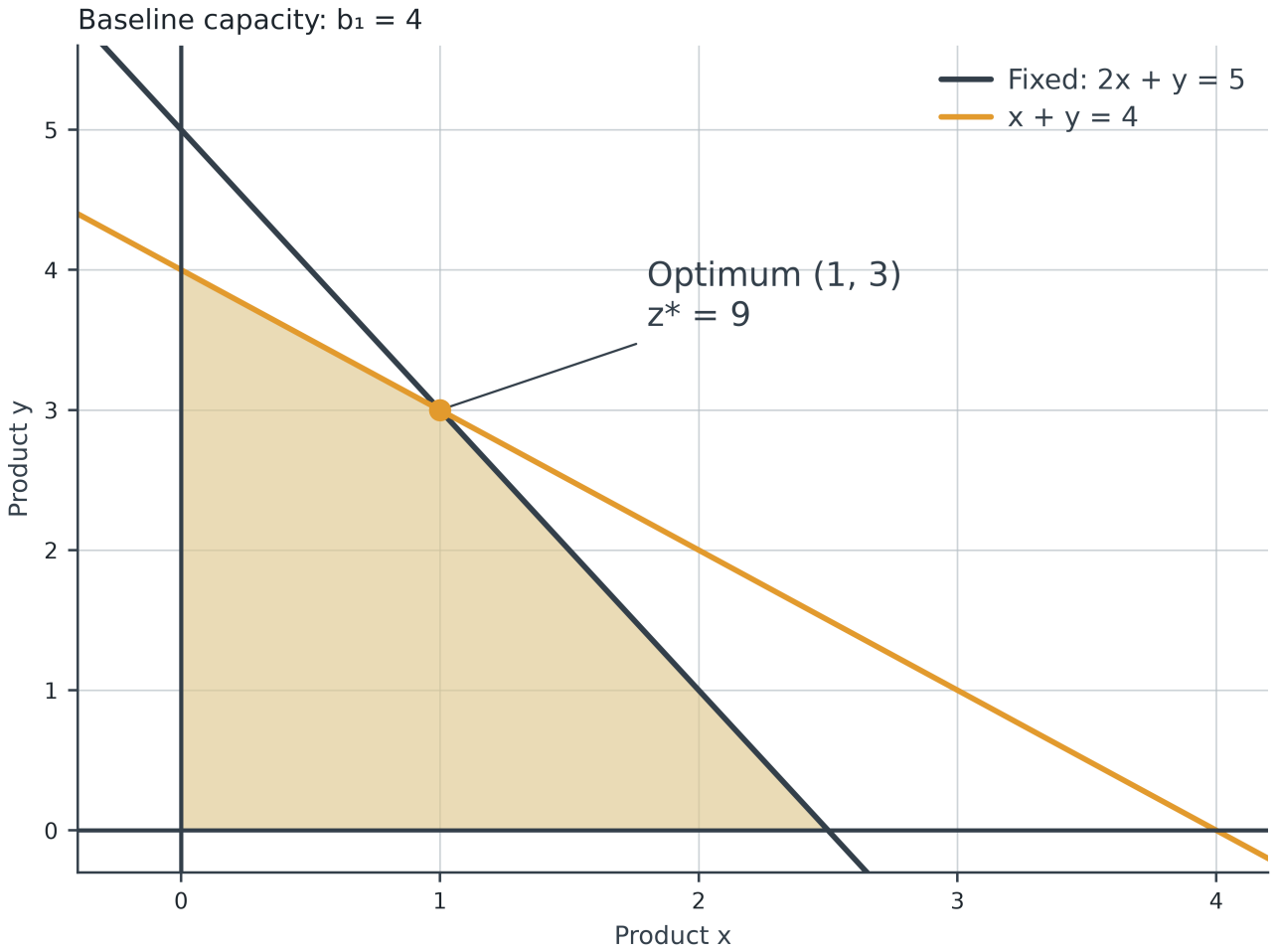
1. Resource 1 changes, resource 2 stays fixed

Maximize $z = 3x + 2y$.

$x + y \le b_1$, $2x + y \le 5$, $x,y \ge 0$.

At $b_1 = 4$: $(x,y) = (1,3)$, $z^* = 9$.

How far can we change b_1 and keep its shadow price equal to 1?



2. The intersection moves with b_1

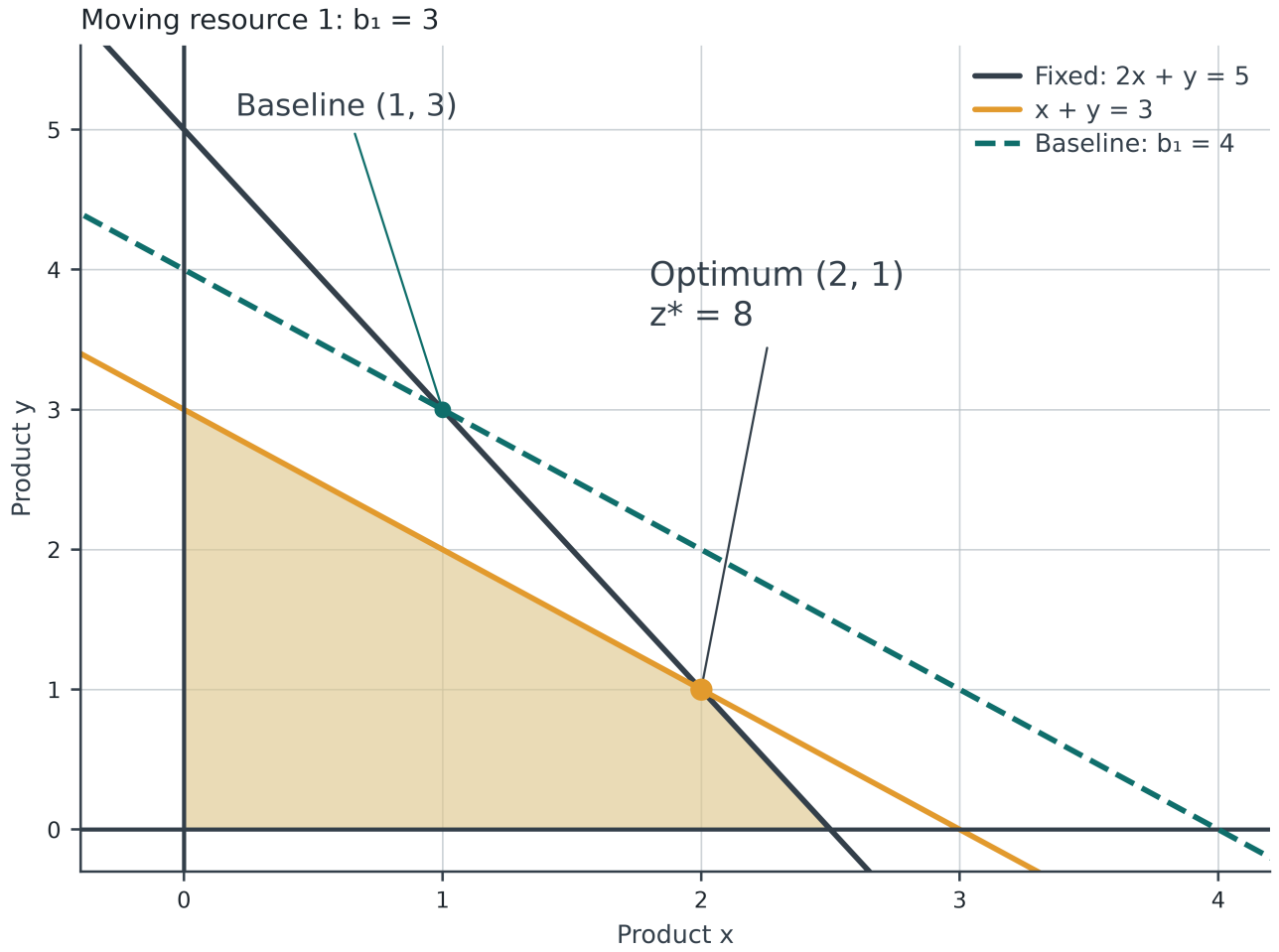
Suppose both resources remain fully used.

$x + y = b_1$

$2x + y = 5$

Subtract the equations: $x = 5 - b_1$.

Substitute back: $y = 2b_1 - 5$.



3. Nonnegativity gives the allowable range

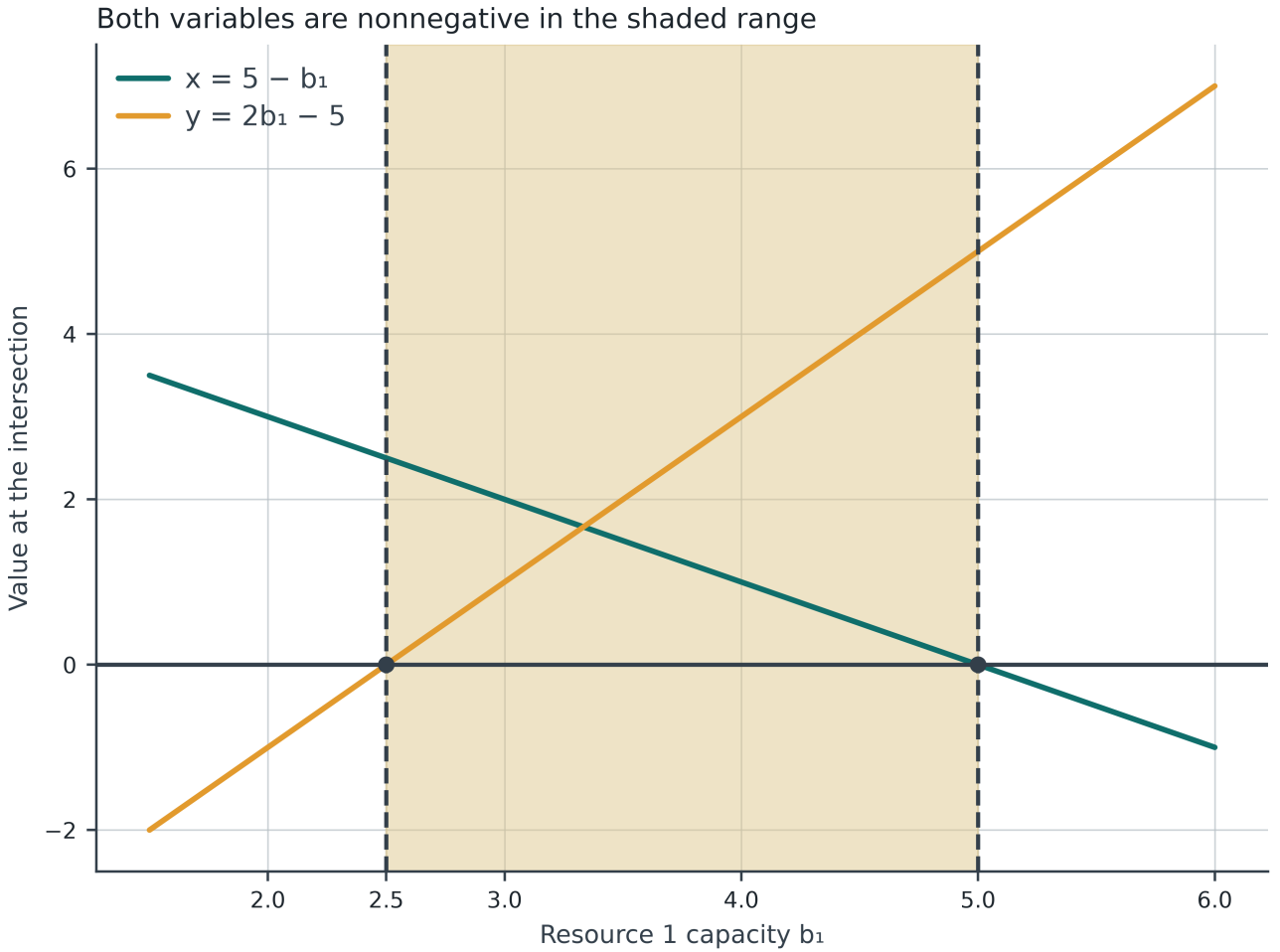
$x = 5 - b_1 \geq 0$ requires $b_1 \leq 5$.

$y = 2b_1 - 5 \geq 0$ requires $b_1 \geq 2.5$.

Together: $2.5 \leq b_1 \leq 5$.

At 2.5: (x,y) = (2.5,0).

At 5: (x,y) = (0,5).



4. Inside the range: $b_1 = 4.5$

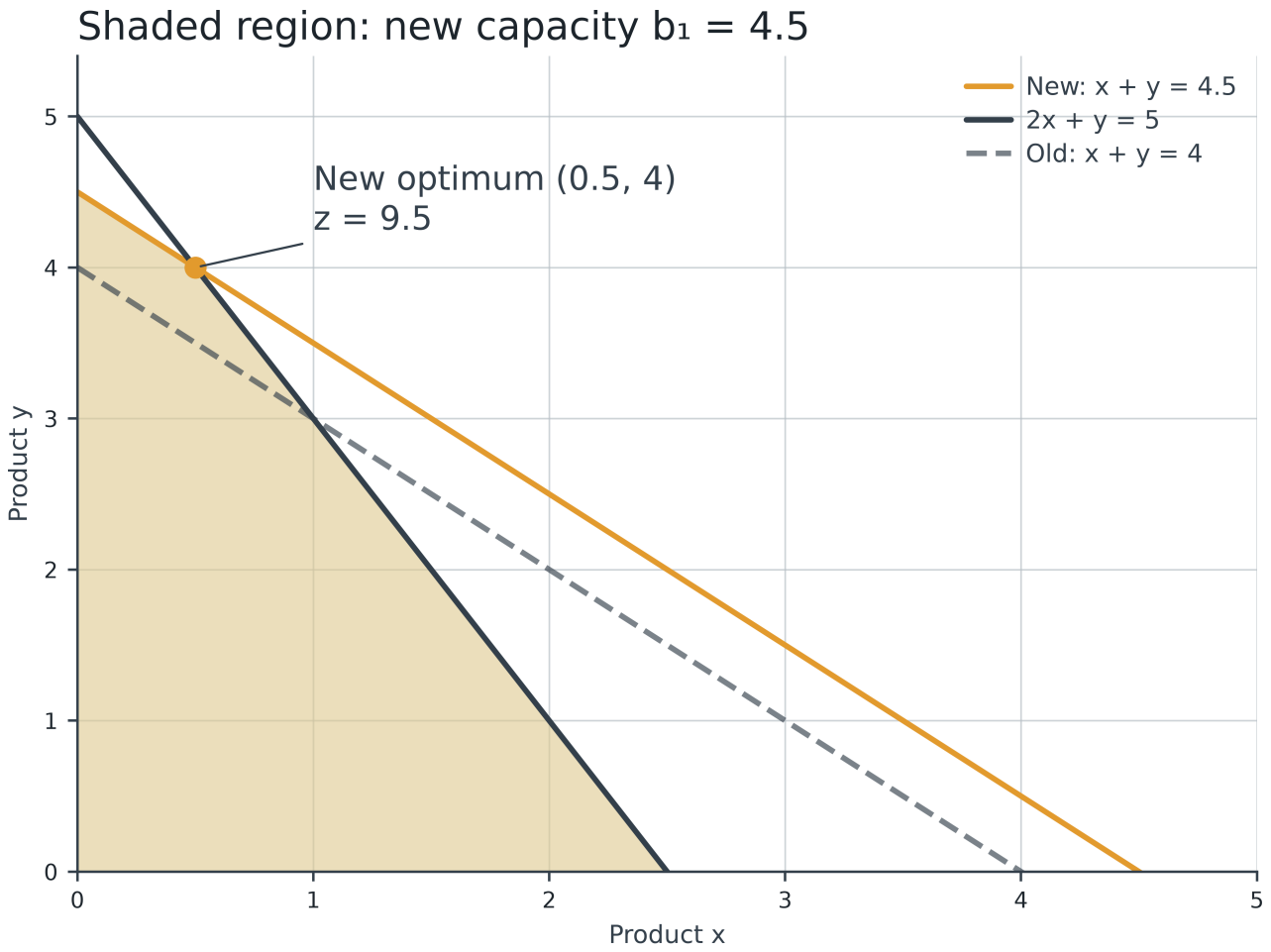
$x = 5 - 4.5 = 0.5$.

$y = 2(4.5) - 5 = 4$.

$z^* = 3(0.5) + 2(4) = 9.5$.

From $b_1 = 4$: $\Delta b_1 = 0.5$, $\Delta z^* = 0.5$.

The extra contribution is \$50.



5. Duality confirms the shadow price

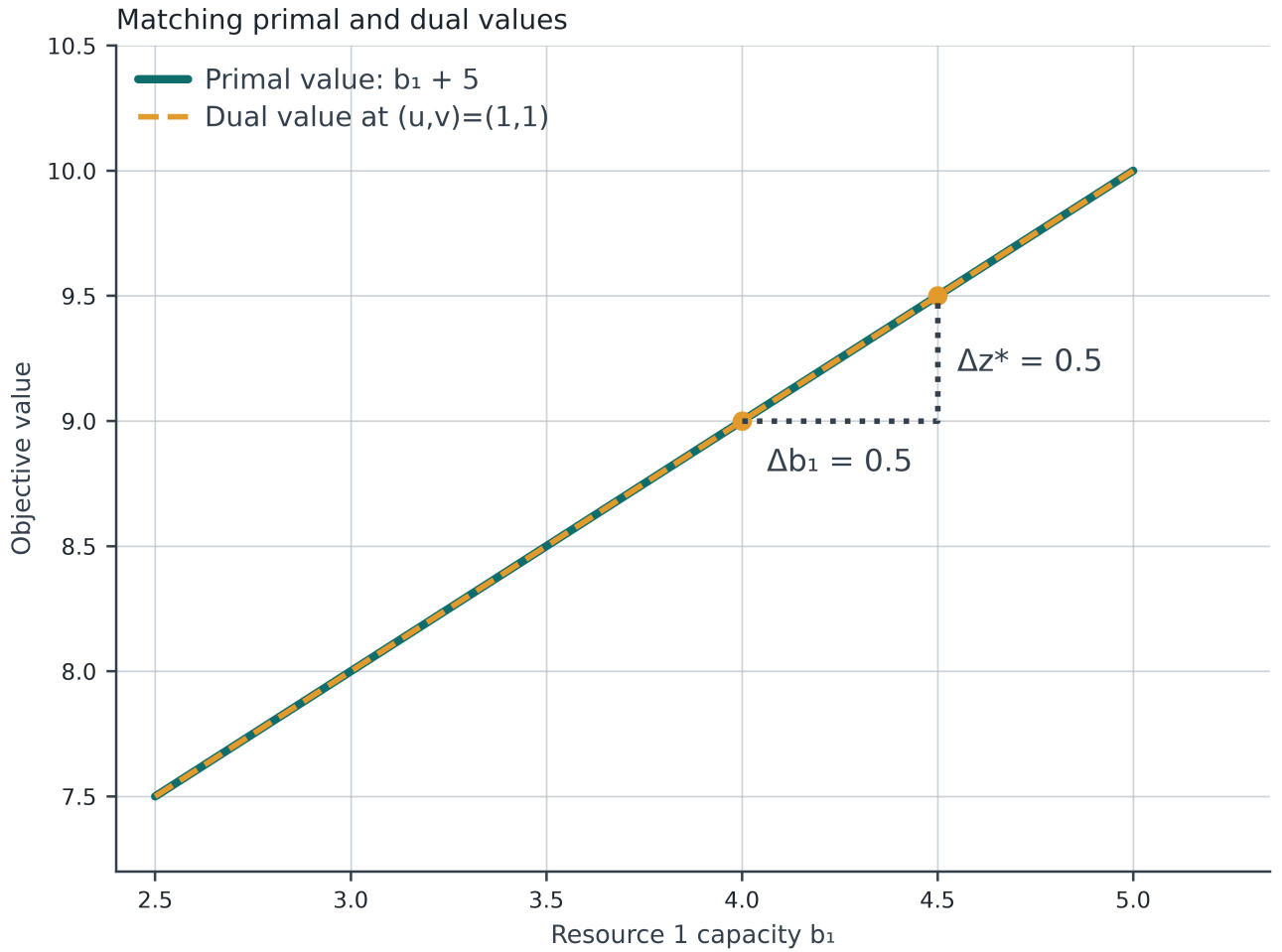
At the intersection: $z^* = 3(5-b_1) + 2(2b_1-5) = b_1 + 5$.

Dual prices (u,v) = (1,1) remain feasible.

Dual value: $w = b_1u + 5v = b_1 + 5$.

Primal value = dual value, so both are optimal.

Within the range: $\Delta z^* = 1 \times \Delta b_1$.



6. Below the range: $b_1 = 2$

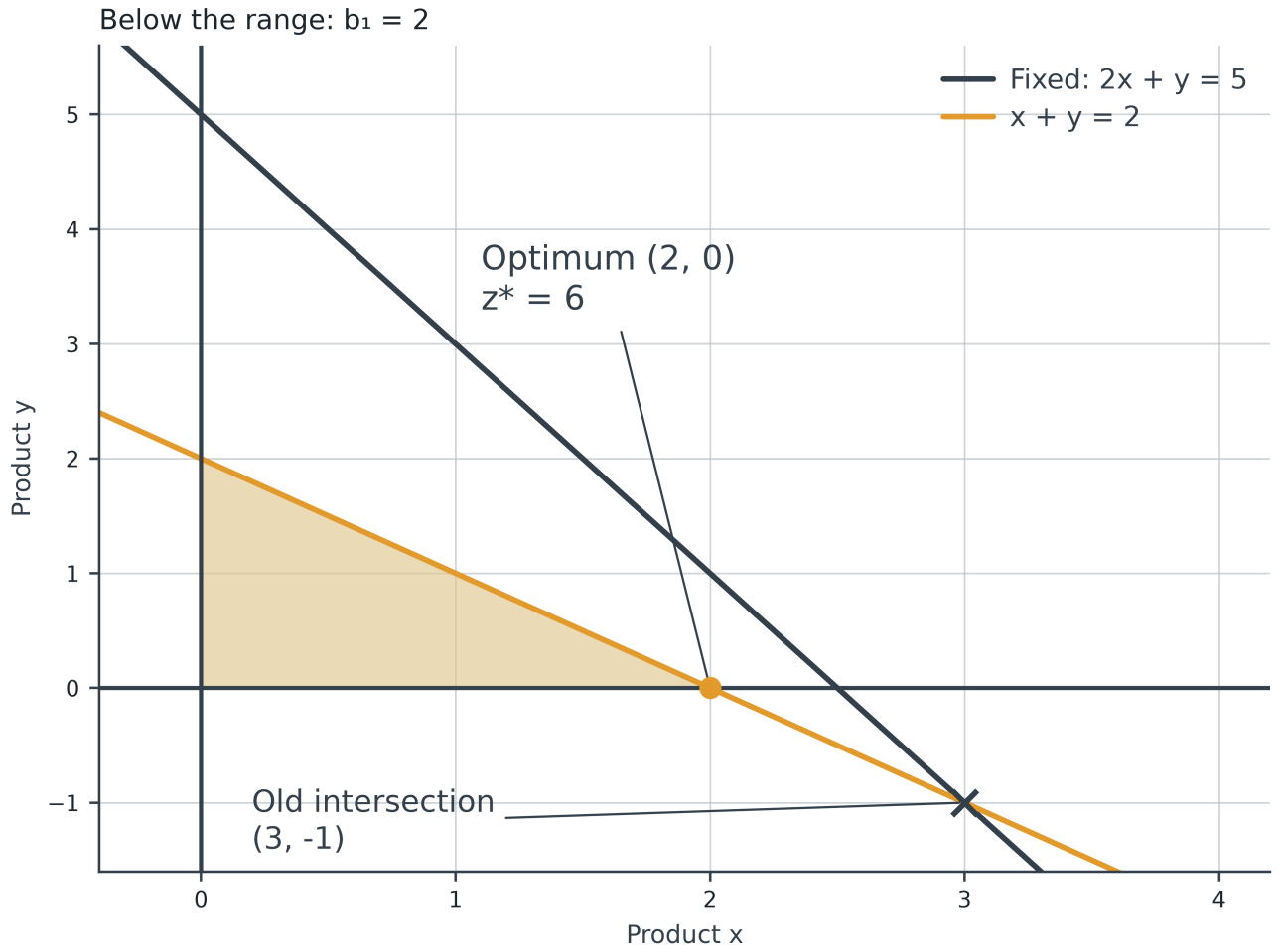
The old intersection is $(3,-1)$, which is infeasible.

The optimum is $(x,y) = (2,0)$, $z^* = 6$.

Resource 2 has slack: $5 - 2(2) = 1$.

For $0 < b_1 < 2.5$: $z^* = 3b_1$.

Shadow price of resource 1: 3.



7. Above the range: $b_1 = 6$

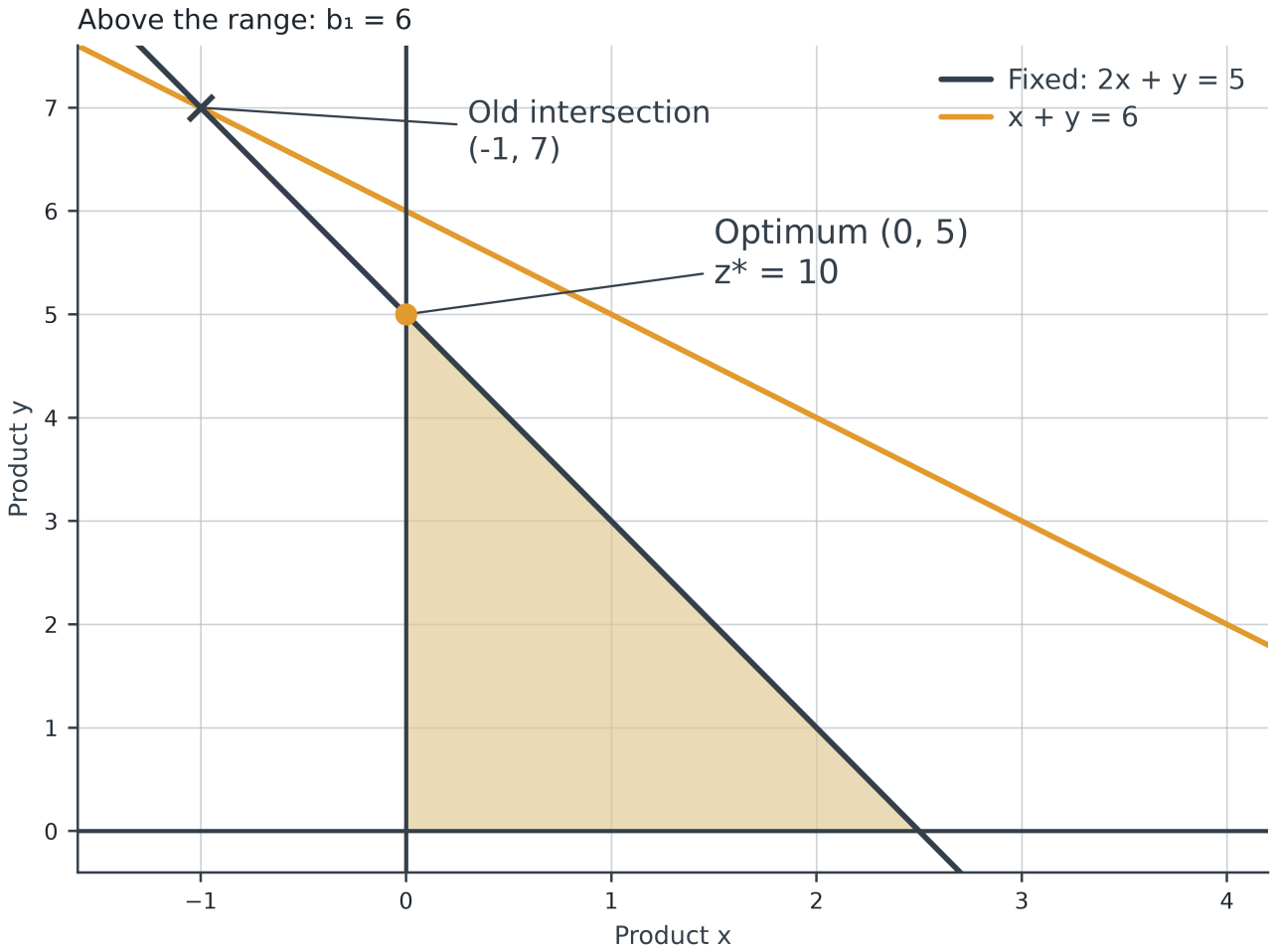
The old intersection is $(-1,7)$, which is infeasible.

The optimum is $(x,y) = (0,5)$, $z^* = 10$.

Resource 1 has slack: $6 - 5 = 1$.

For $b_1 > 5$: $z^* = 10$.

Shadow price of resource 1: 0.



8. The value function changes slope

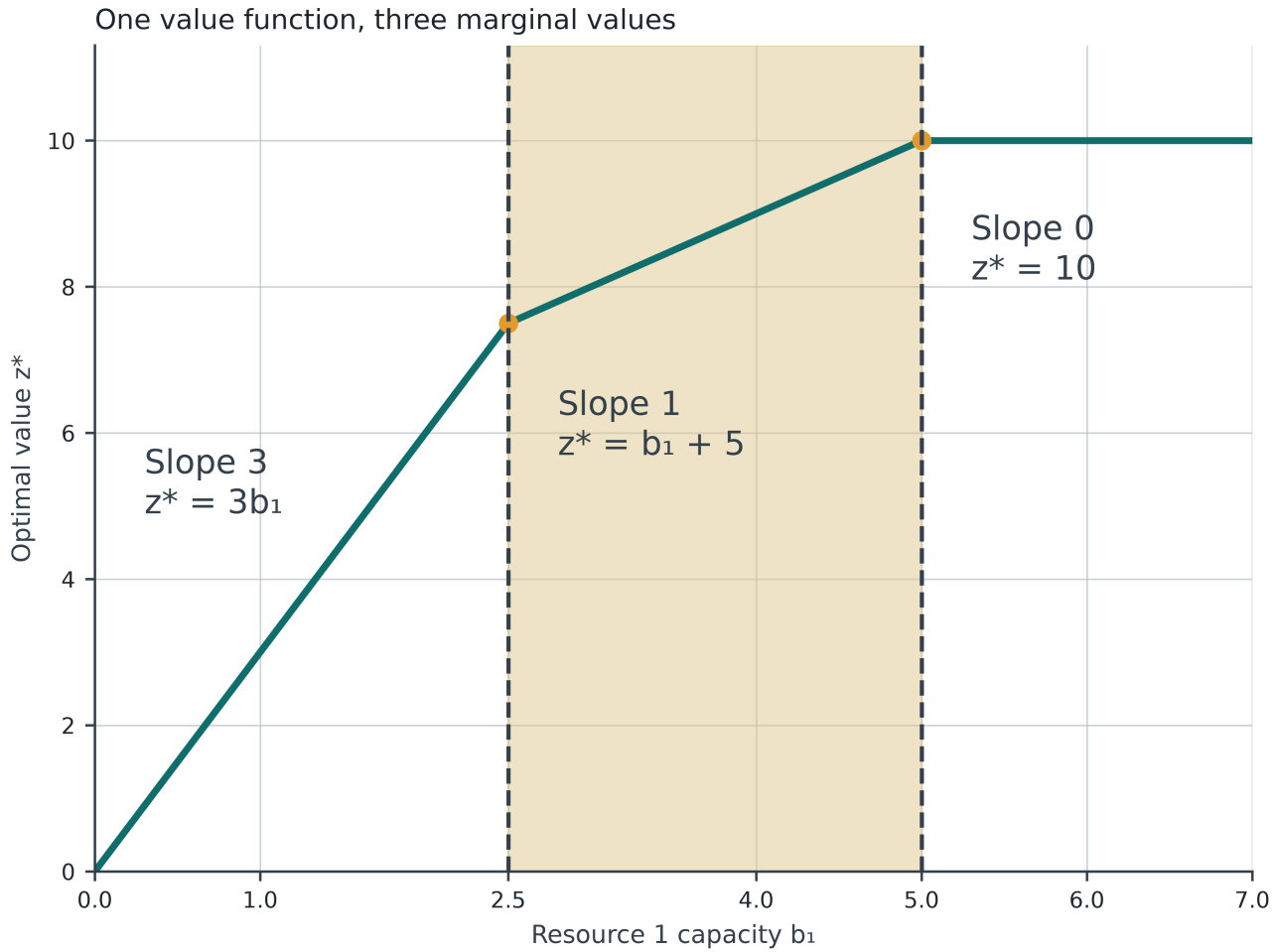
$0 \le b_1 \le 2.5$: $z^* = 3b_1$.

$2.5 \le b_1 \le 5$: $z^* = b_1 + 5$.

$b_1 \ge 5$: $z^* = 10$.

The middle formula includes both endpoints.

At 2.5 and 5, left and right slopes differ.



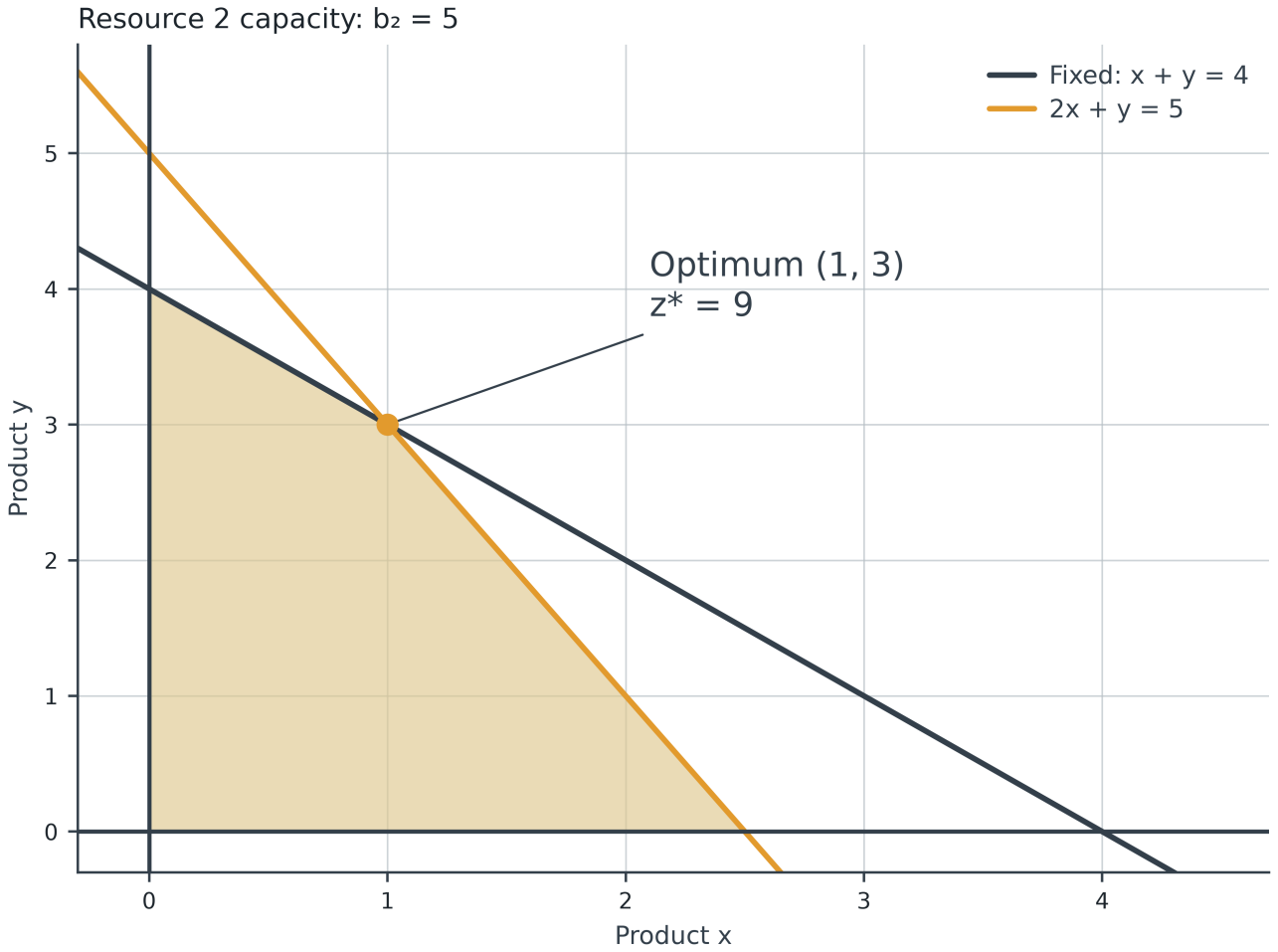
1. Resource 2 changes, resource 1 stays fixed

Maximize $z = 3x + 2y$.

$x + y \le 4$, $2x + y \le b_2$, $x,y \ge 0$.

At $b_2 = 5$: $(x,y) = (1,3)$, $z^* = 9$.

How far can b_2 change while its shadow price stays at 1?



2. The intersection as a function of b_2

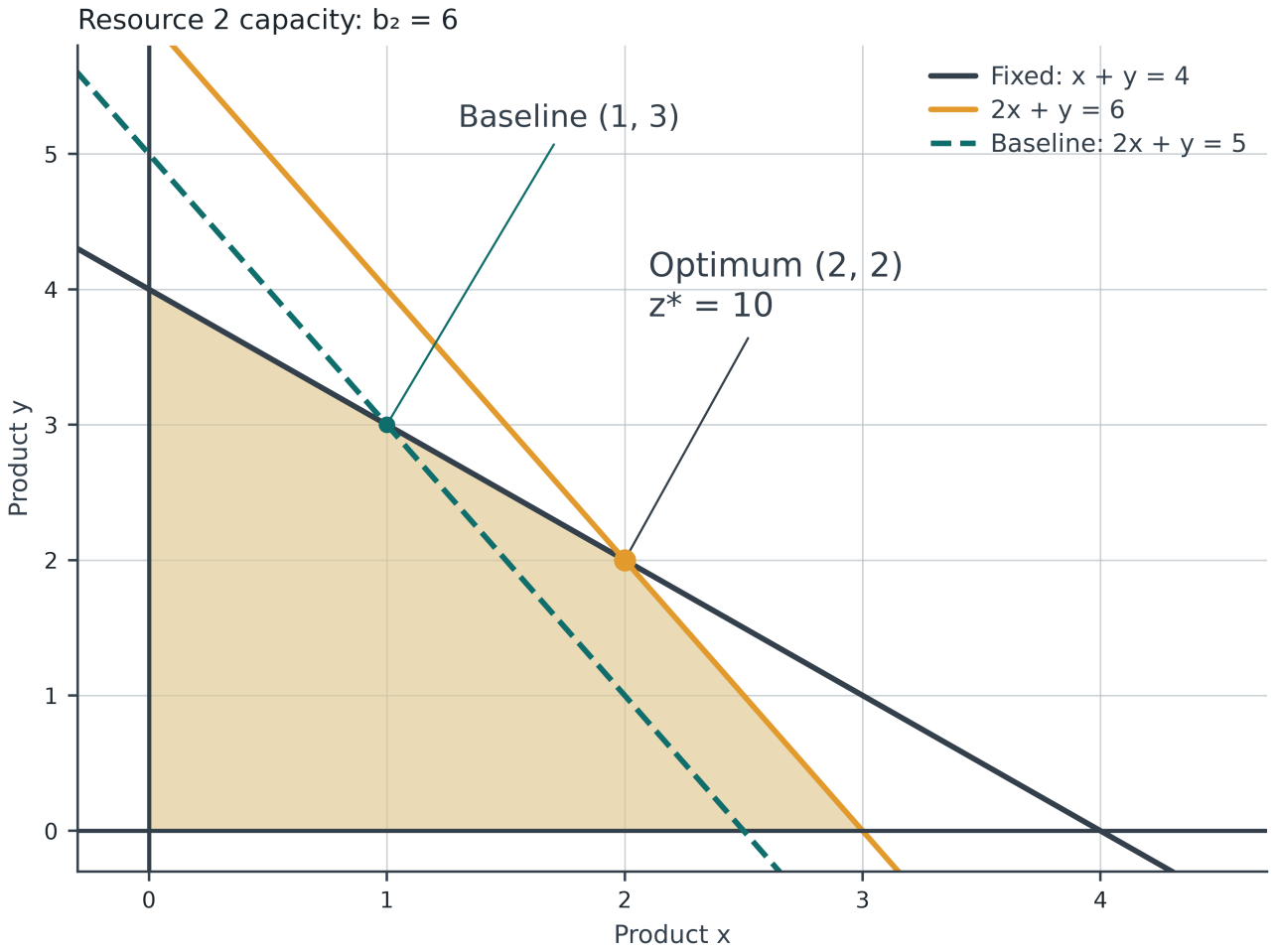
Suppose both resources remain fully used.

$x + y = 4$

$2x + y = b_2$

Subtract the equations: $x = b_2 - 4$.

Substitute back: $y = 8 - b_2$.



3. Nonnegativity gives $4 \le b_2 \le 8$

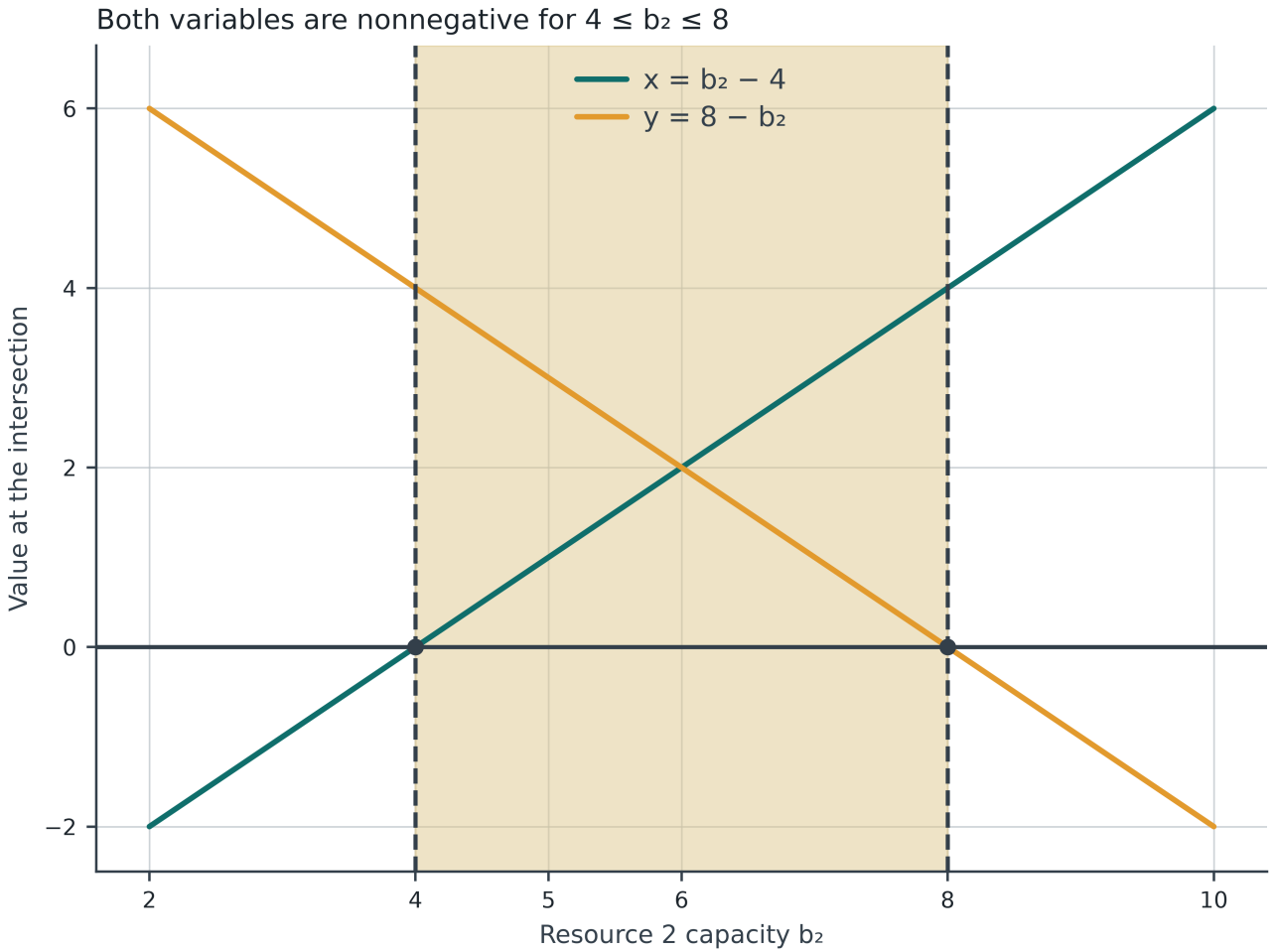
$x = b_2 - 4 \ge 0$ requires $b_2 \ge 4$.

$y = 8 - b_2 \ge 0$ requires $b_2 \le 8$.

Together: $4 \le b_2 \le 8$.

At 4: (x,y) = (0,4). At 8: (x,y) = (4,0).

From $b_2 = 5$: decrease by 1 or increase by 3.



4. Inside the range: $b_2 = 6$

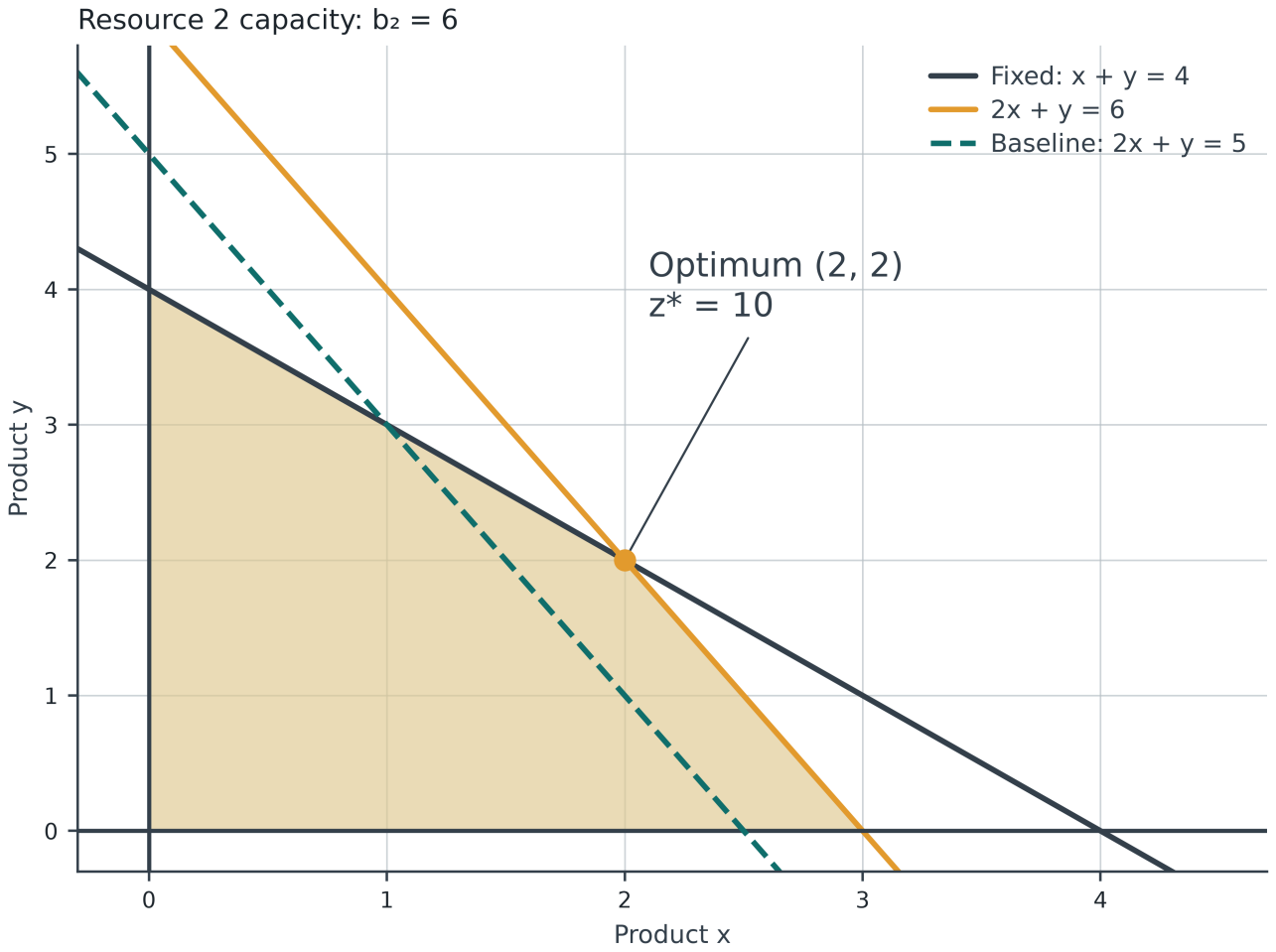
$x = 6 - 4 = 2.$

$y = 8 - 6 = 2.$

$z^* = 3(2) + 2(2) = 10.$

From $b_2 = 5$: $\Delta b_2 = 1$, $\Delta z^* = 1$.

The extra contribution is \$100.



5. Duality confirms the price of resource 2

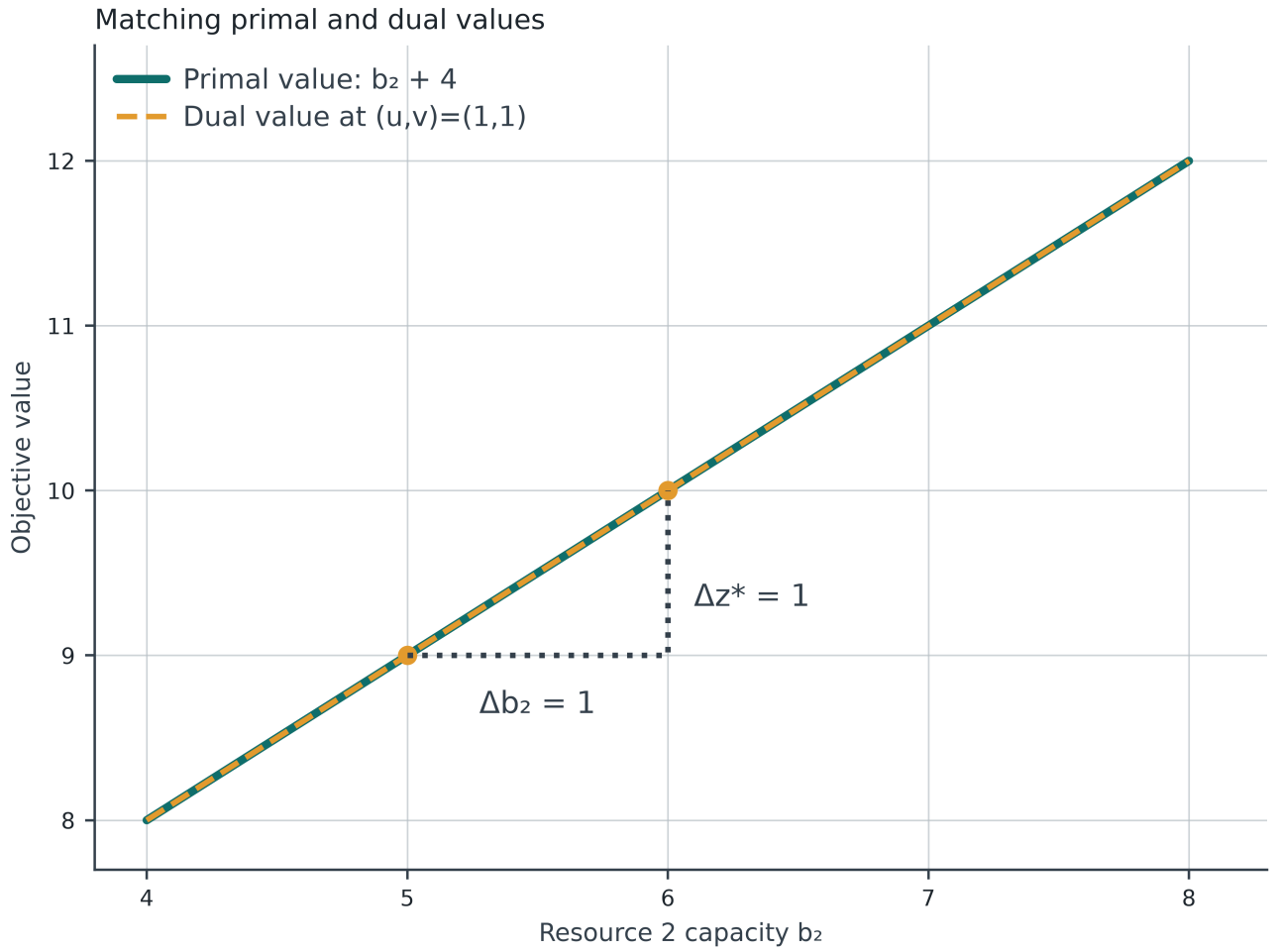
At the intersection: $z^* = 3(b_2-4) + 2(8-b_2) = b_2 + 4$.

Dual prices $(u,v) = (1,1)$ remain feasible.

Dual value: $w = 4u + b_2v = b_2 + 4$.

Primal value = dual value, so both are optimal.

Within the range: $\Delta z^* = 1 \times \Delta b_2$.



6. Below the range: $b_2 = 3$

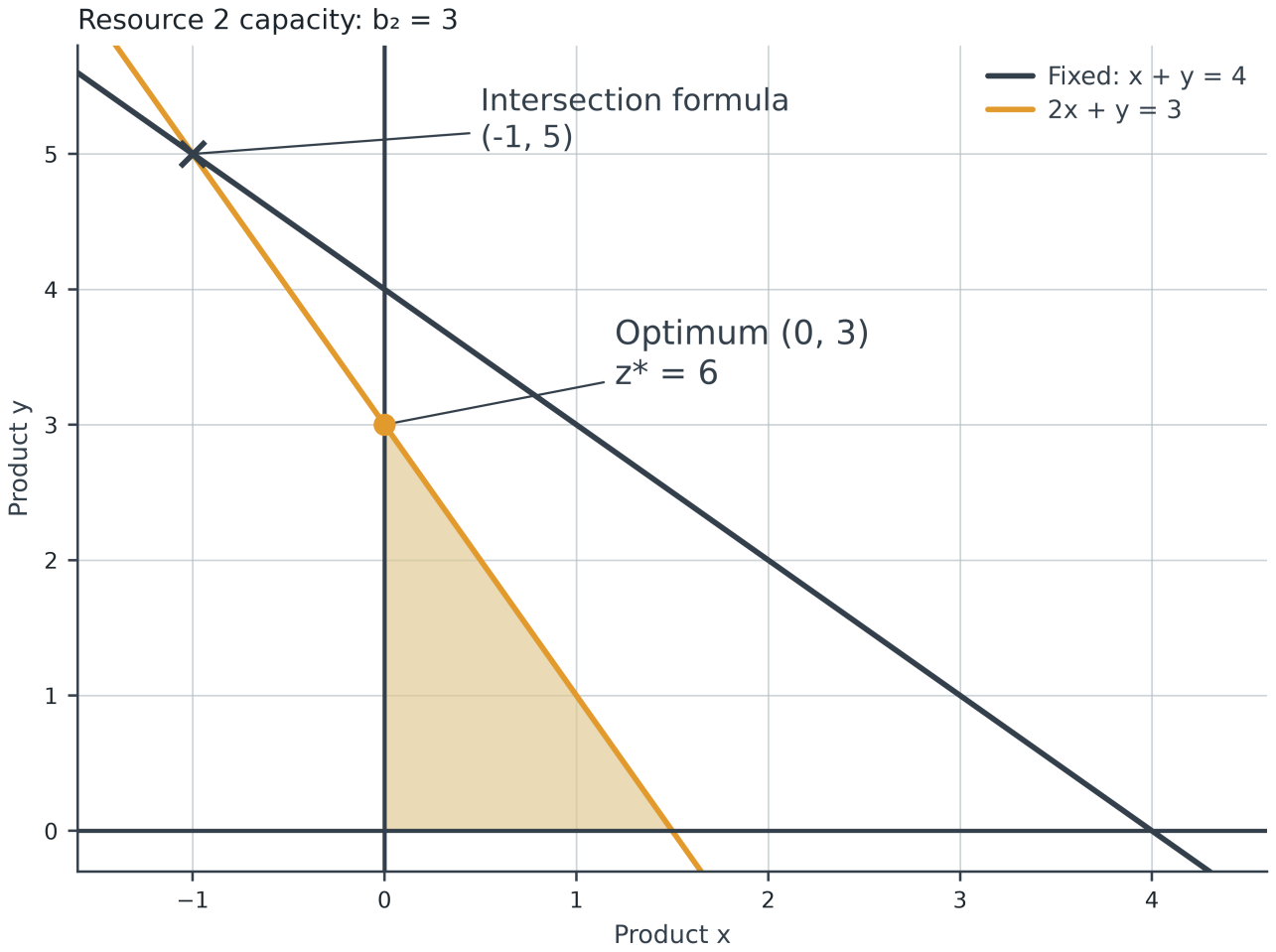
The intersection formula gives $(-1,5)$, which is infeasible.

The optimum is $(x,y) = (0,3)$, $z^* = 6$.

Resource 1 has slack: $4 - 3 = 1$.

For $0 < b_2 < 4$: $z^* = 2b_2$.

Shadow price of resource 2: 2.



7. Above the range: $b_2 = 9$

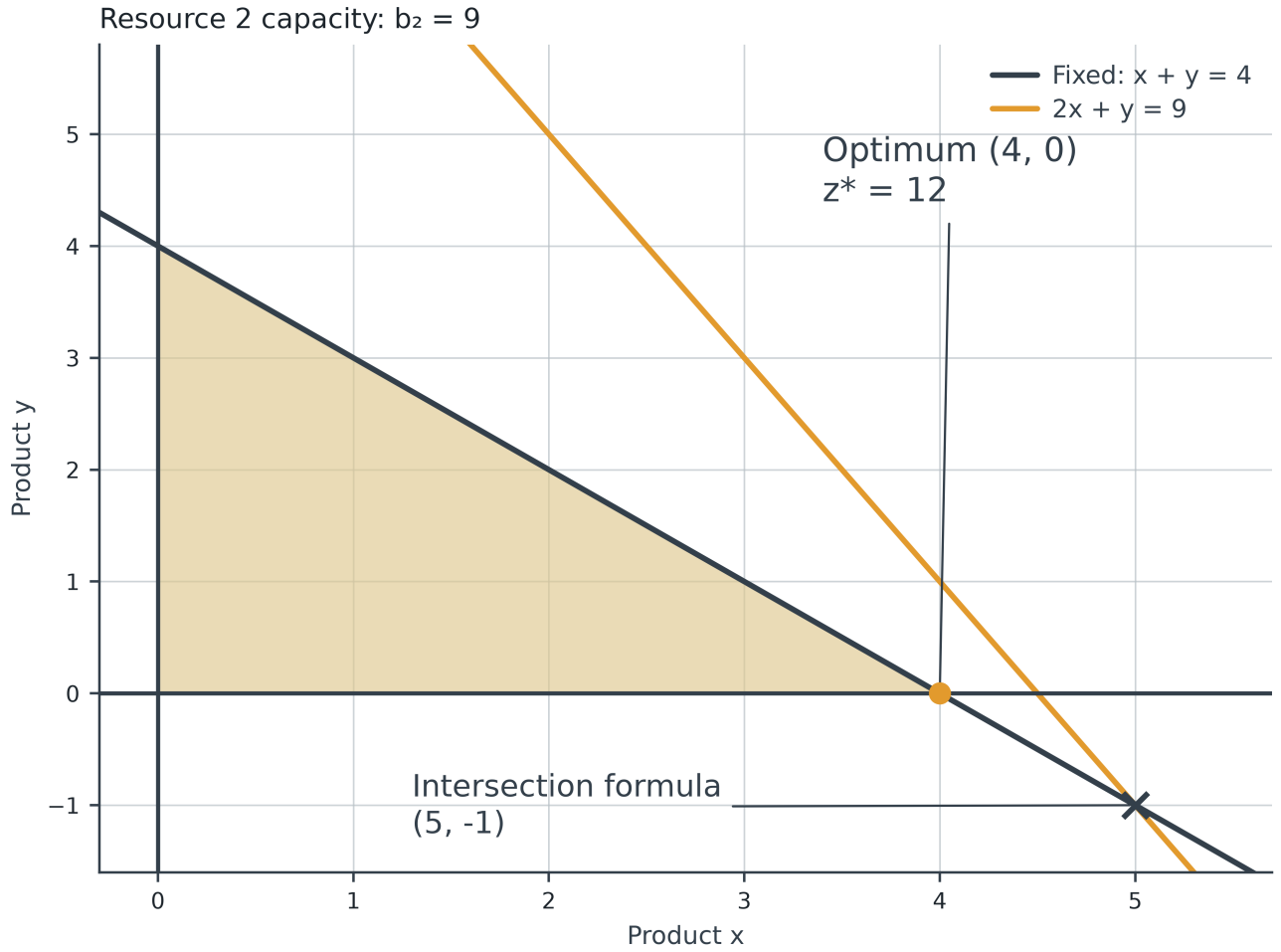
The intersection formula gives $(5,-1)$, which is infeasible.

The optimum is $(x,y) = (4,0)$, $z^* = 12$.

Resource 2 has slack: $9 - 2(4) = 1$.

For $b_2 > 8$: $z^* = 12$.

Shadow price of resource 2: 0.



8. The value function for resource 2

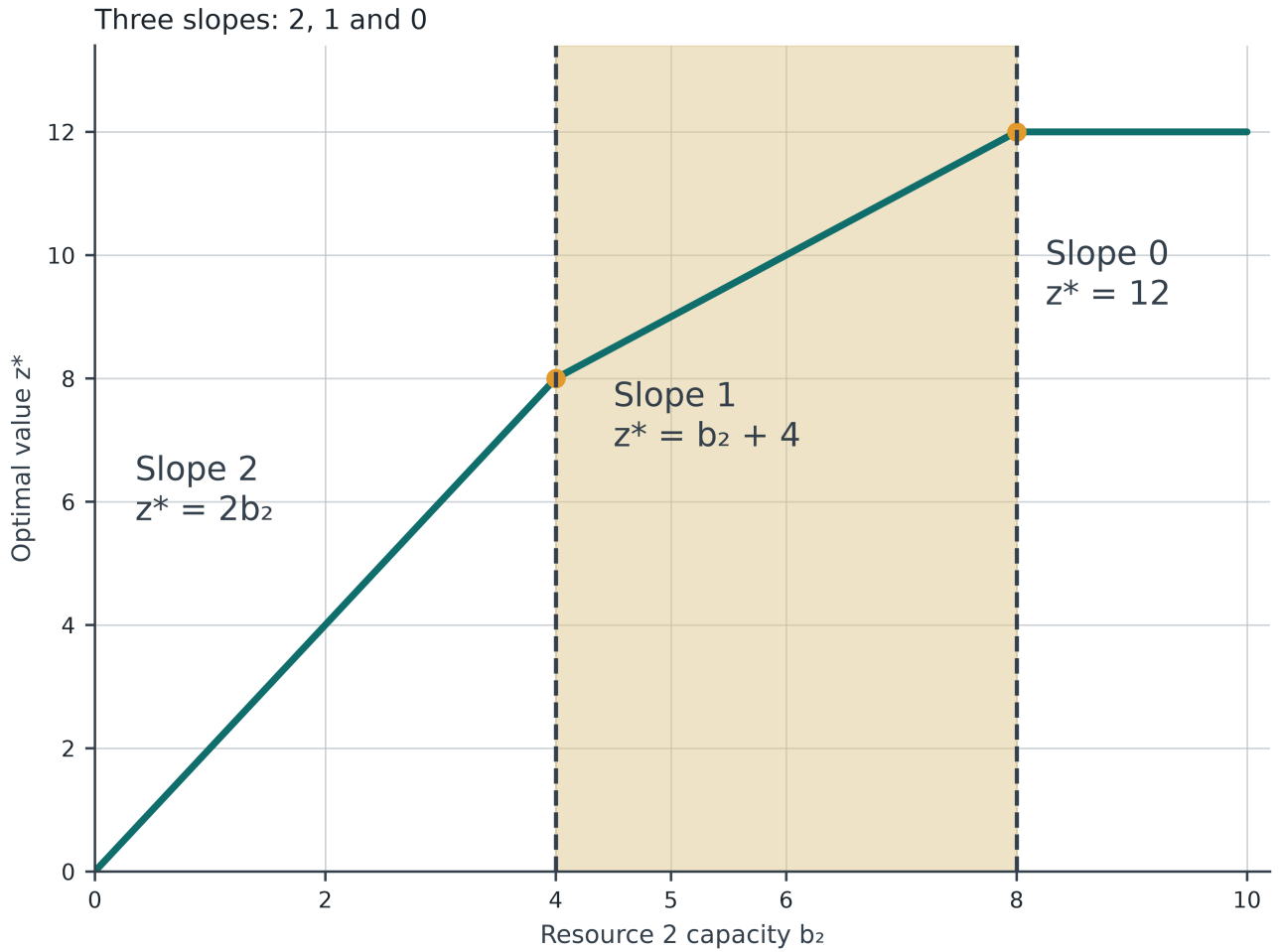
$0 \le b_2 \le 4$: $z^* = 2b_2$.

$4 \le b_2 \le 8$: $z^* = b_2 + 4$.

$b_2 \ge 8$: $z^* = 12$.

At 4: left slope 2, right slope 1.

At 8: left slope 1, right slope 0.



9. Changing both resources needs a joint check

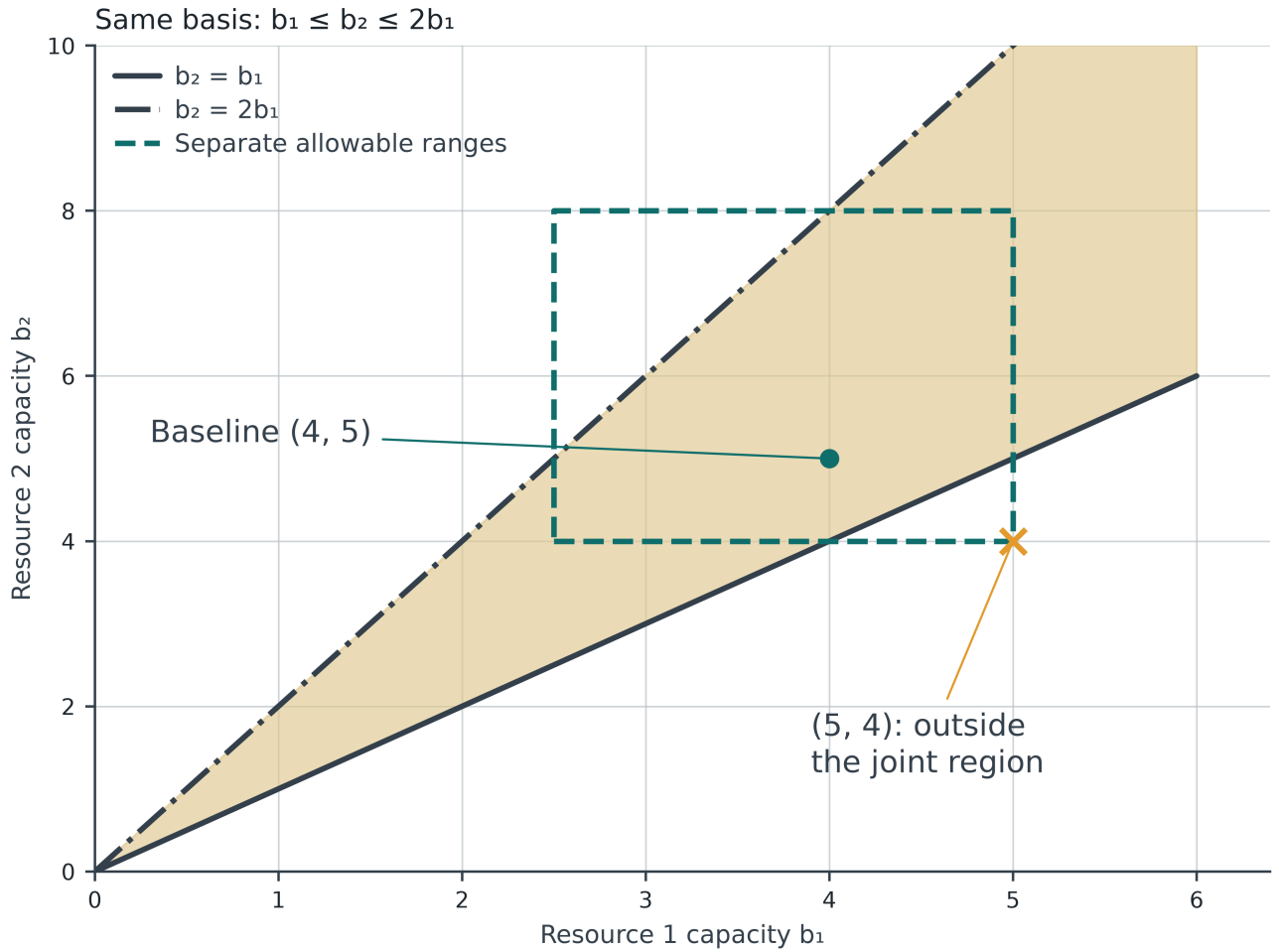
With both RHS values changing:

$x = b_2 - b_1$, $y = 2b_1 - b_2$.

The same basis requires $b_1 \le b_2 \le 2b_1$.

$(b_1,b_2) = (5,4)$ is outside this region.

Its true optimum is $(x,y)=(0,4)$, $z^*=8$.



1. Objective sensitivity keeps the feasible set fixed

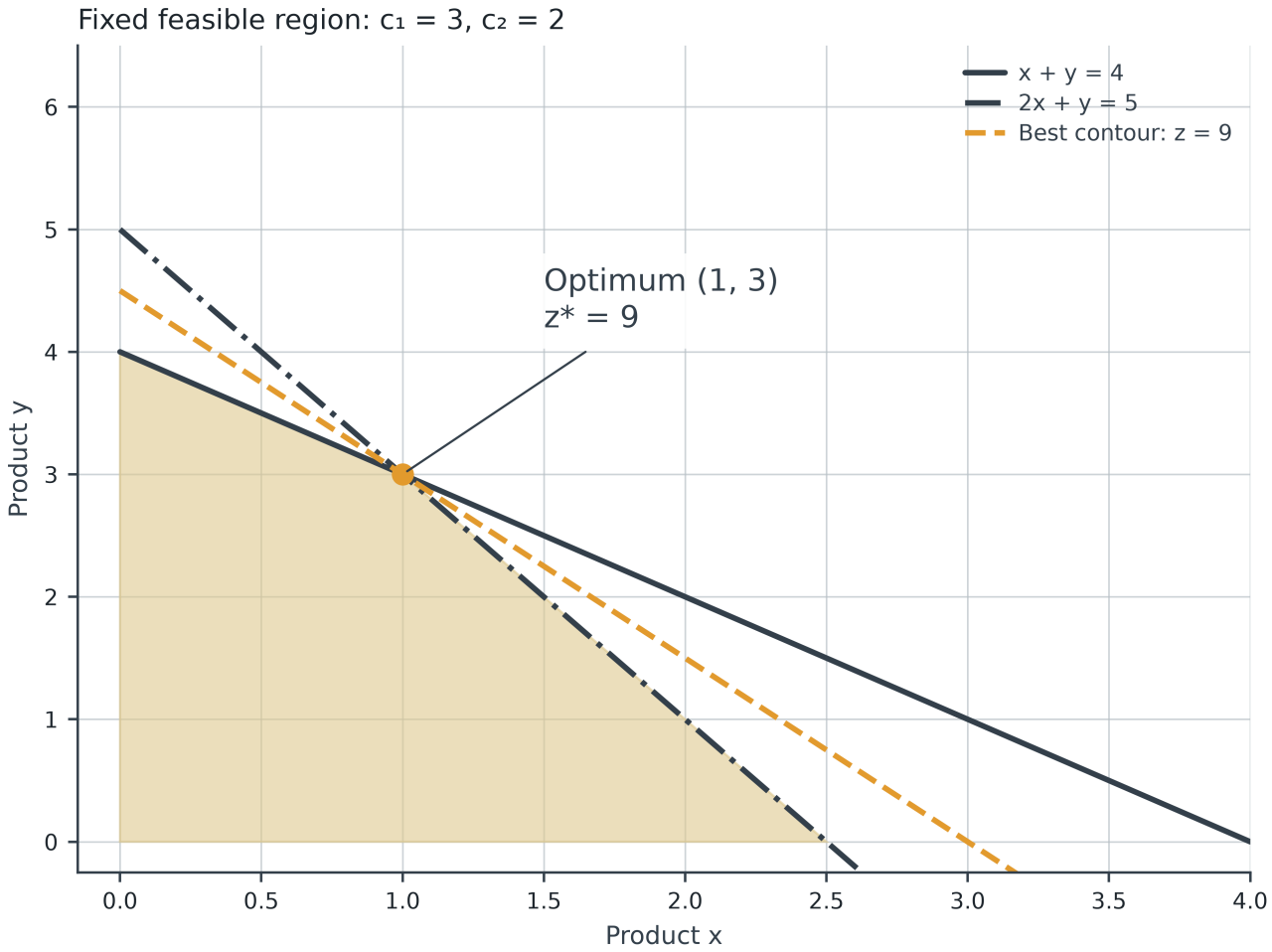
Maximize $z = c_1x + 2y$.

$x + y \le 4$, $2x + y \le 5$, $x,y \ge 0$.

Baseline $c_1 = 3$: $(x,y) = (1,3)$, $z^* = 9$.

Only the contribution from product x changes.

Which c_1 values keep (1,3) optimal?



2. Complementary slackness determines the prices

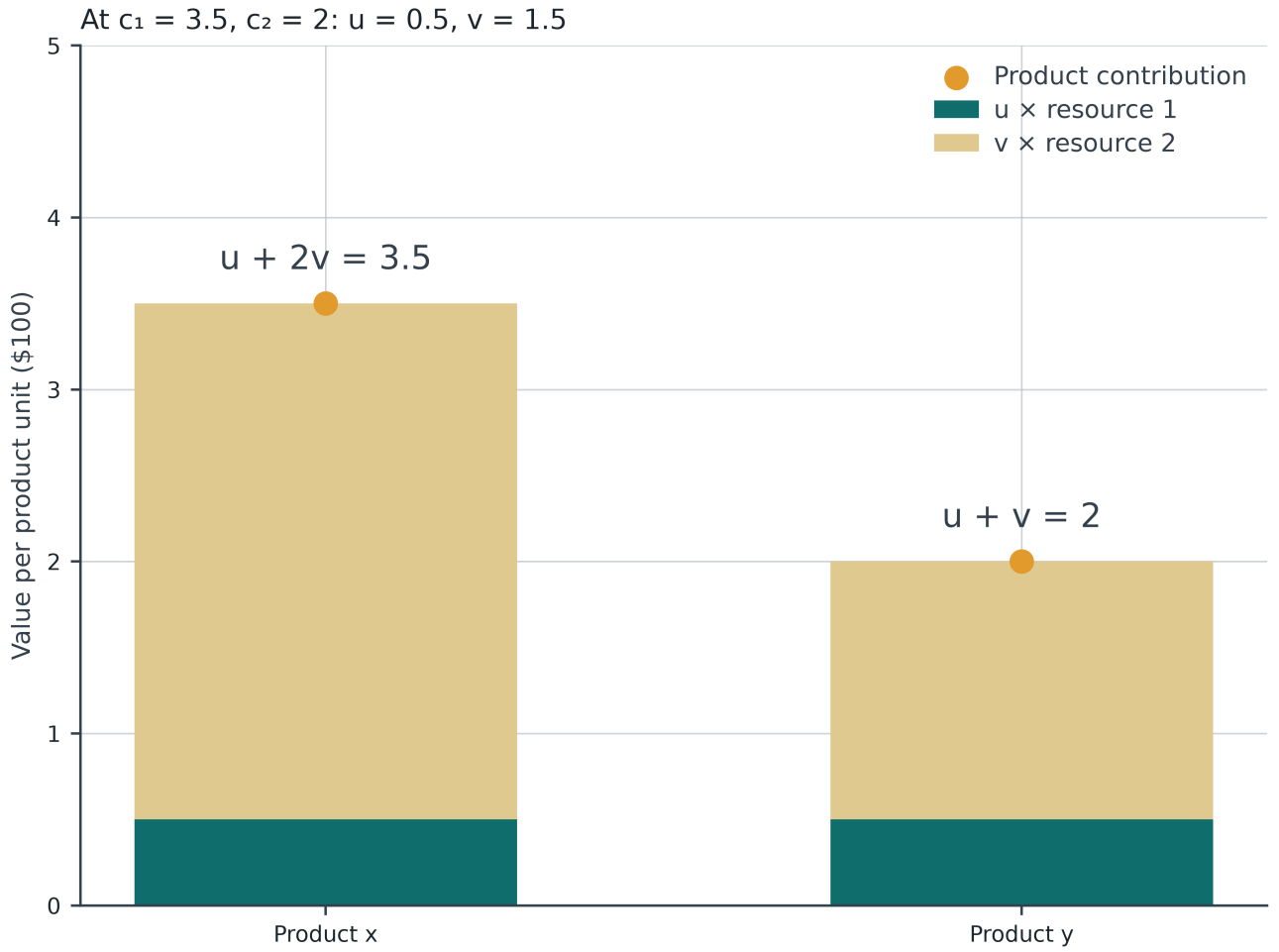
At (1,3), both products have positive output.

Their dual constraints must be tight:

$u + 2v = c_1$, $u + v = c_2$.

Subtract: $v = c_1 - c_2$.

Then $u = 2c_2 - c_1$.



3. Nonnegative dual prices give the range

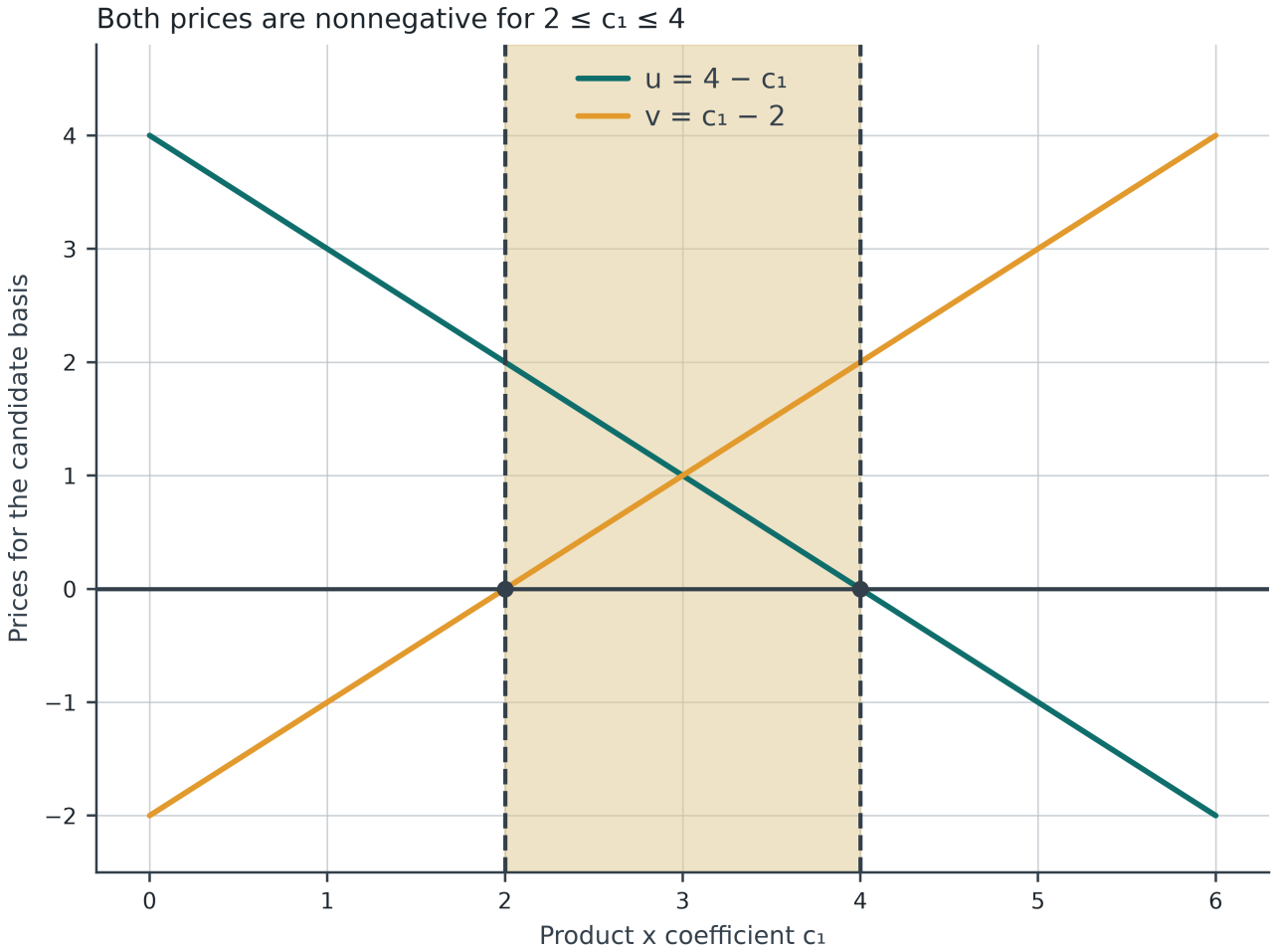
$u = 2c_2 - c_1 \geq 0$ requires $c_1 \leq 2c_2$.

$v = c_1 - c_2 \geq 0$ requires $c_1 \geq c_2$.

Together: $c_2 \leq c_1 \leq 2c_2$.

With $c_2 = 2$: $2 \leq c_1 \leq 4$.

The contour slope $-c_1/2$ lies between -2 and -1.



4. Below the range: $c_1 = 1$

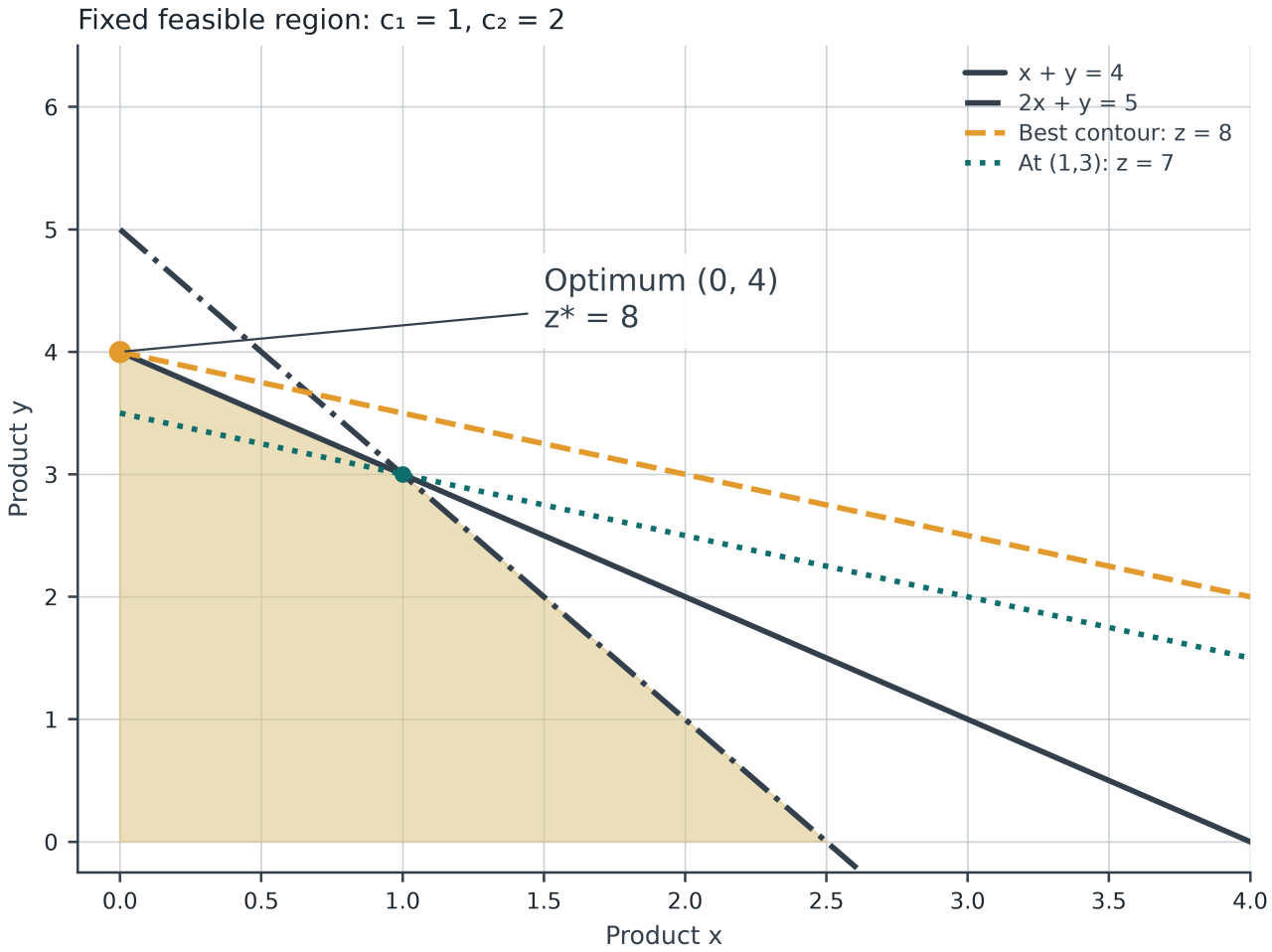
Maximize $z = x + 2y$.

At (1,3): $z = 7$.

The optimum is (0,4), with $z^* = 8$.

The old price formula gives $v = -1$.

For $c_1 < 2$: (0,4) stays optimal.



5. At $c_1 = 2$, an entire edge is optimal

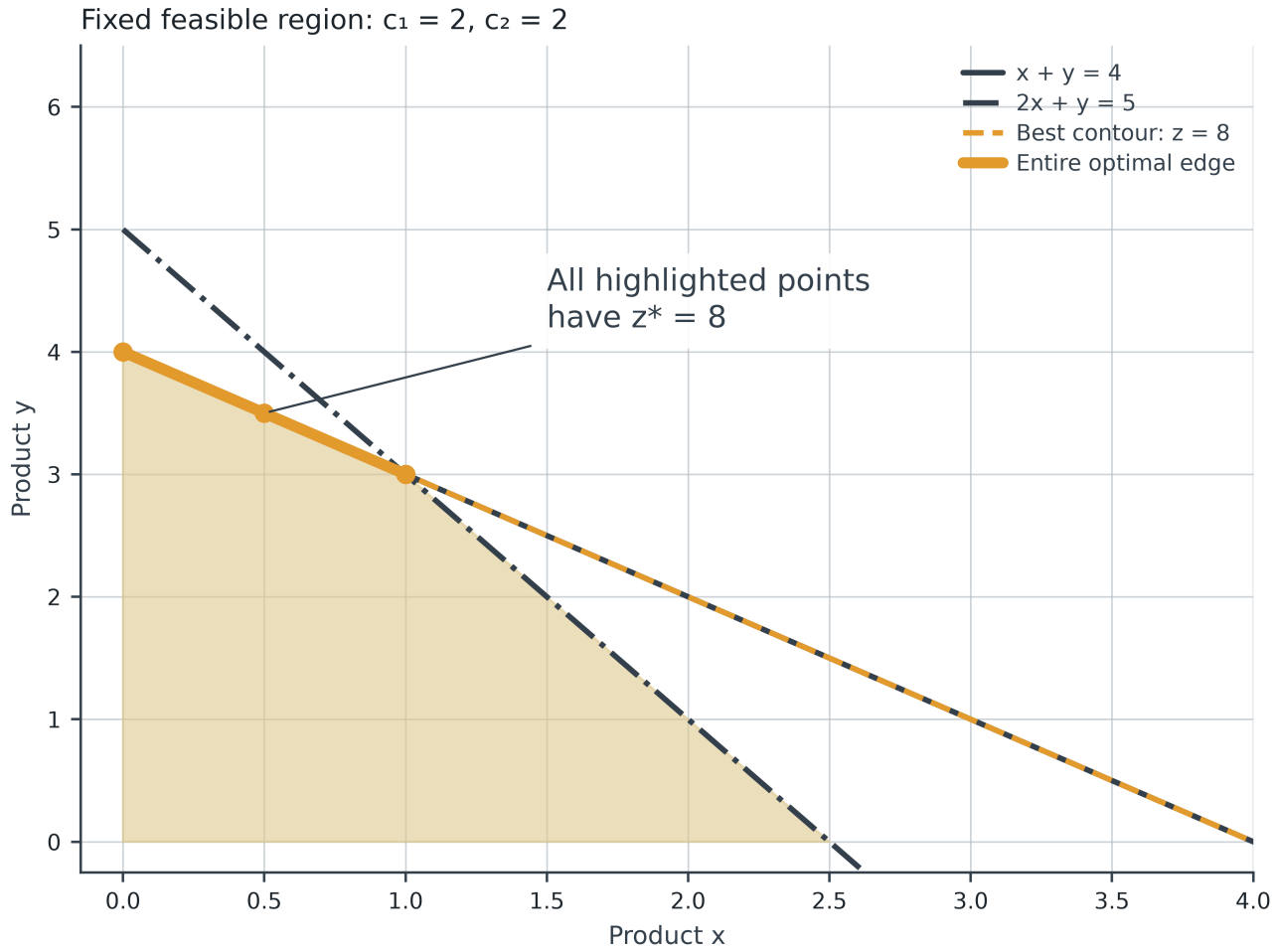
$z = 2x + 2y = 2(x+y) \le 8$.

Every feasible point on x+y=4 attains 8.

The optimal edge runs from (0,4) to (1,3).

(1,3) is still optimal, but not unique.

Dual prices: (u,v) = (2,0).



6. Inside the range: $c_1 = 3.5$

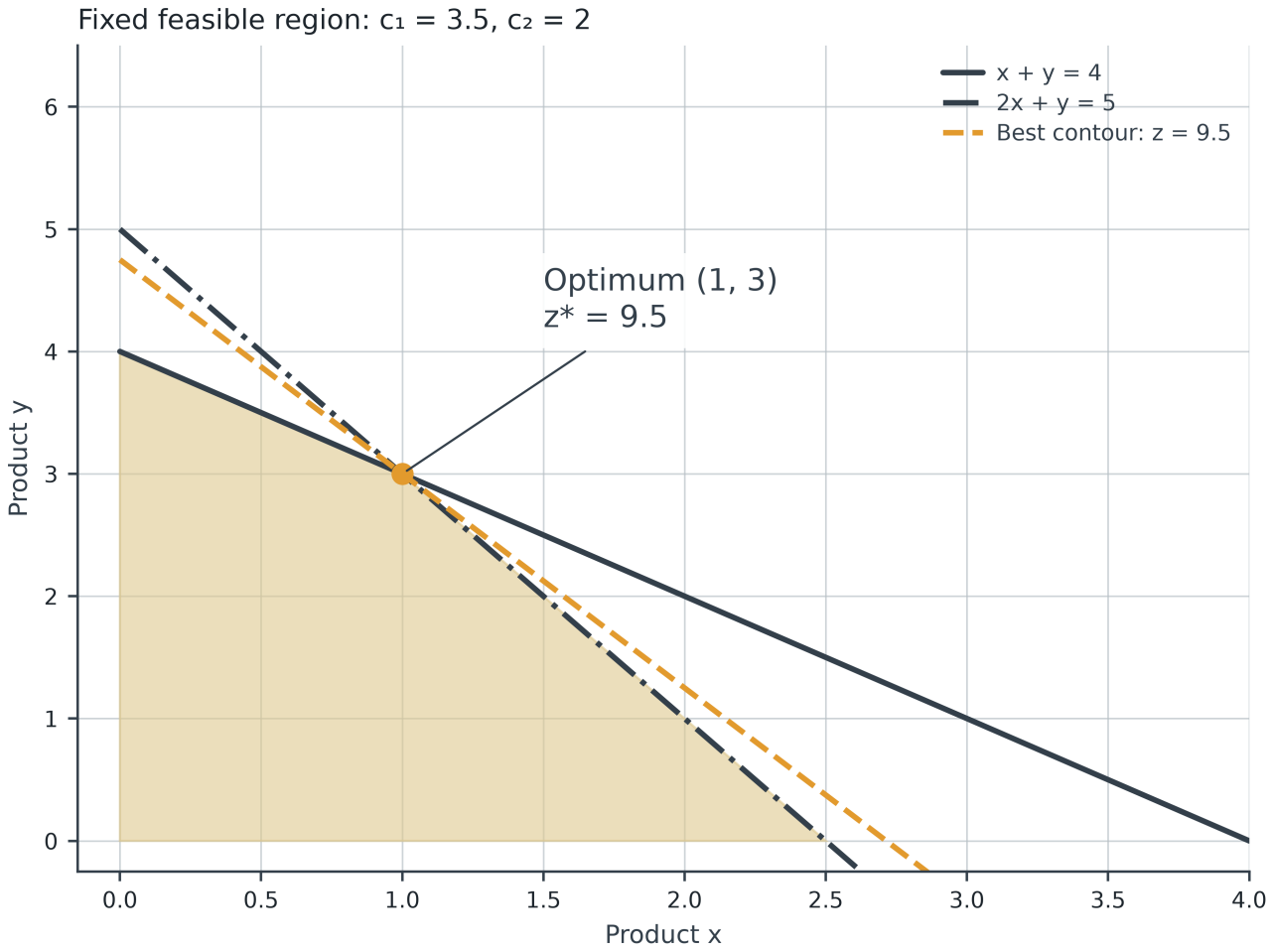
The optimum stays at (x,y) = (1,3).

$z^* = 3.5(1) + 2(3) = 9.5$.

$u = 4 - 3.5 = 0.5$, $v = 3.5 - 2 = 1.5$.

Dual value: $4u + 5v = 9.5$.

The production plan stays fixed, but resource prices change.



7. At $c_1 = 4$, the other edge is optimal

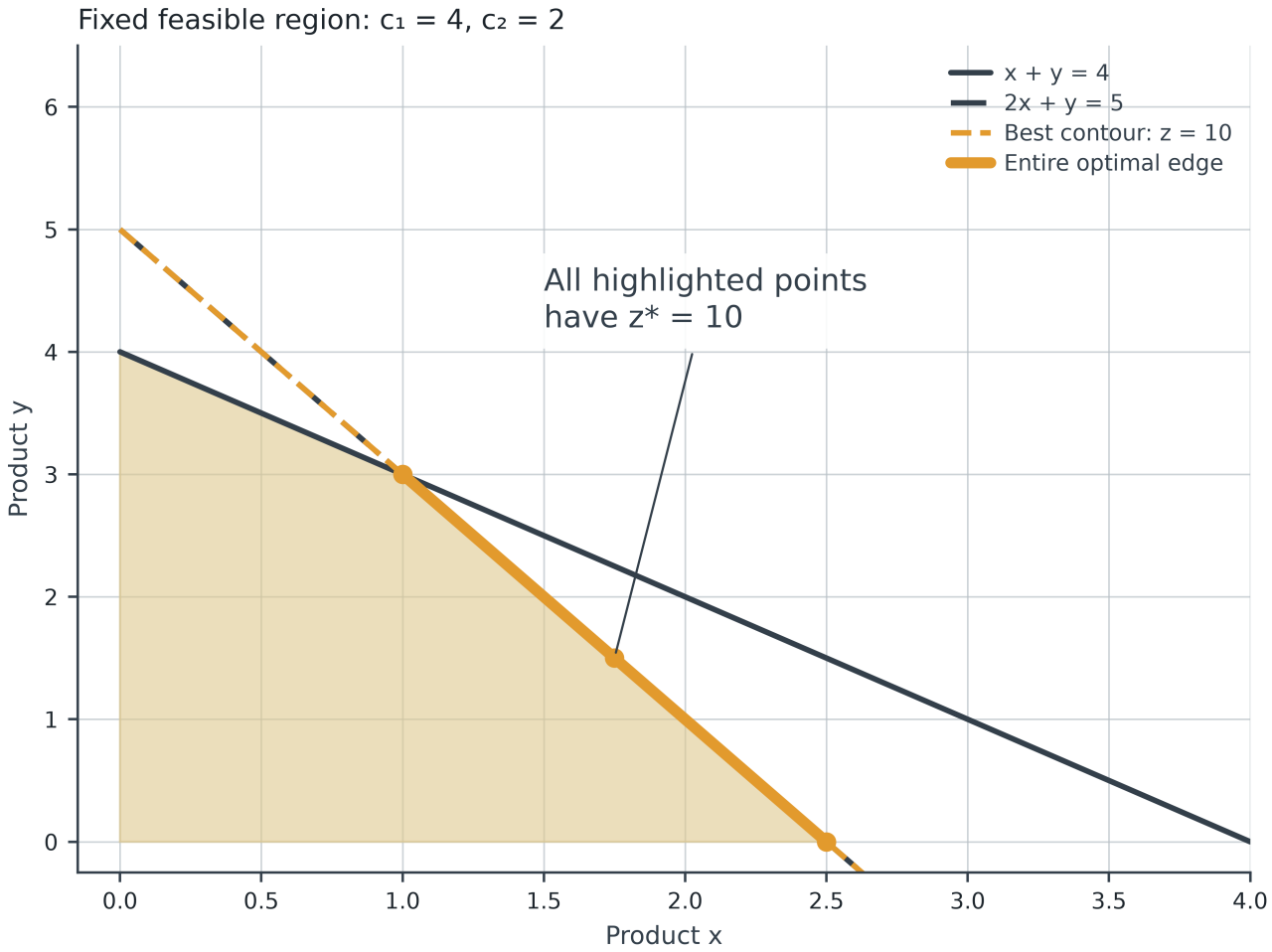
$z = 4x + 2y = 2(2x+y) \le 10.$

Every feasible point on 2x+y=5 attains 10.

The optimal edge runs from (1,3) to (2.5,0).

(1,3) remains one of the optimal solutions.

Dual prices: (u,v) = (0,2).



8. Above the range: $c_1 = 5$

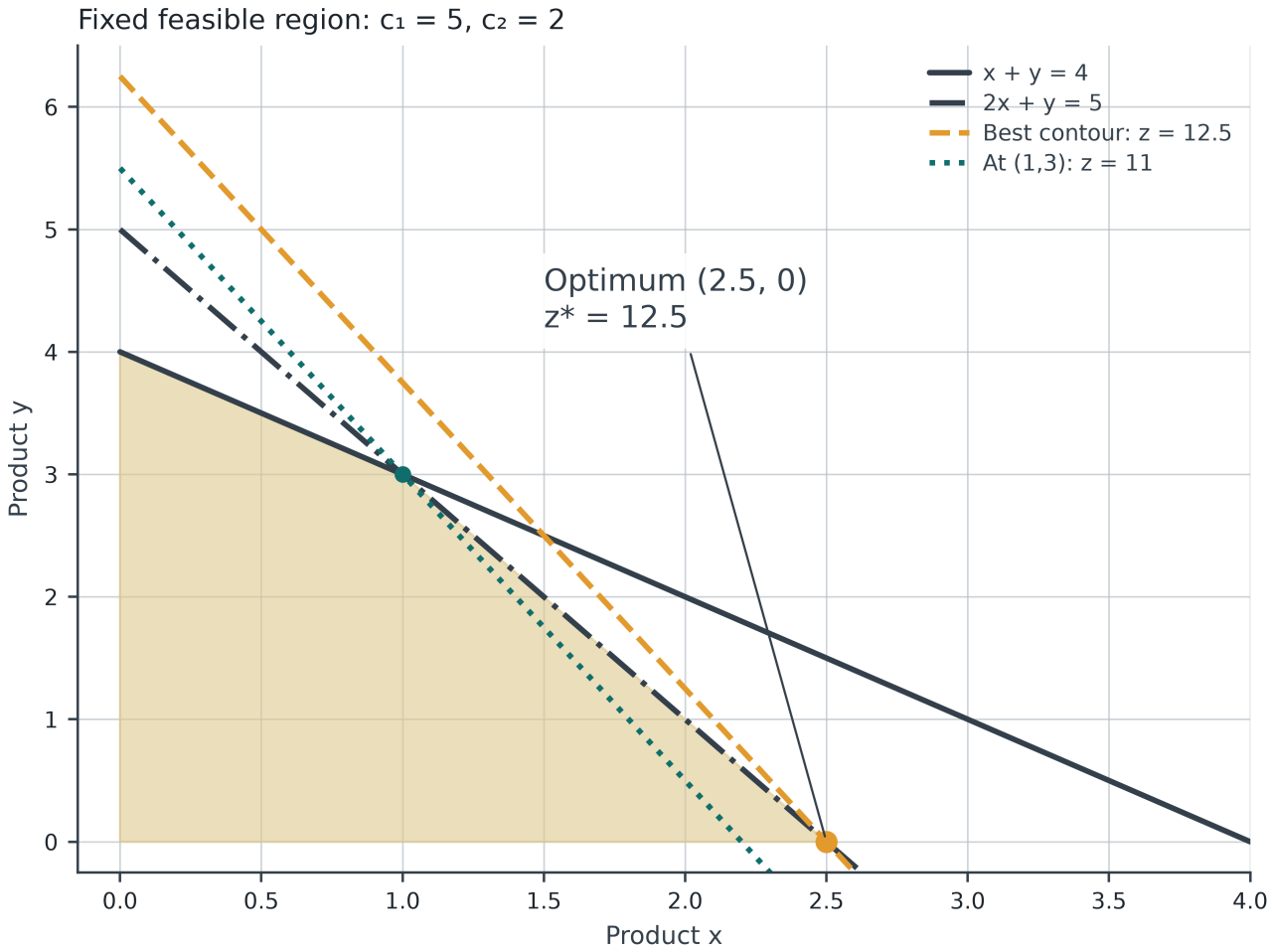
Maximize $z = 5x + 2y$.

At (1,3): $z = 11$.

The optimum is (2.5,0), with $z^* = 12.5$.

The old price formula gives $u = -1$.

For $c_1 > 4$: (2.5,0) stays optimal.



9. Optimal value as the coefficient changes

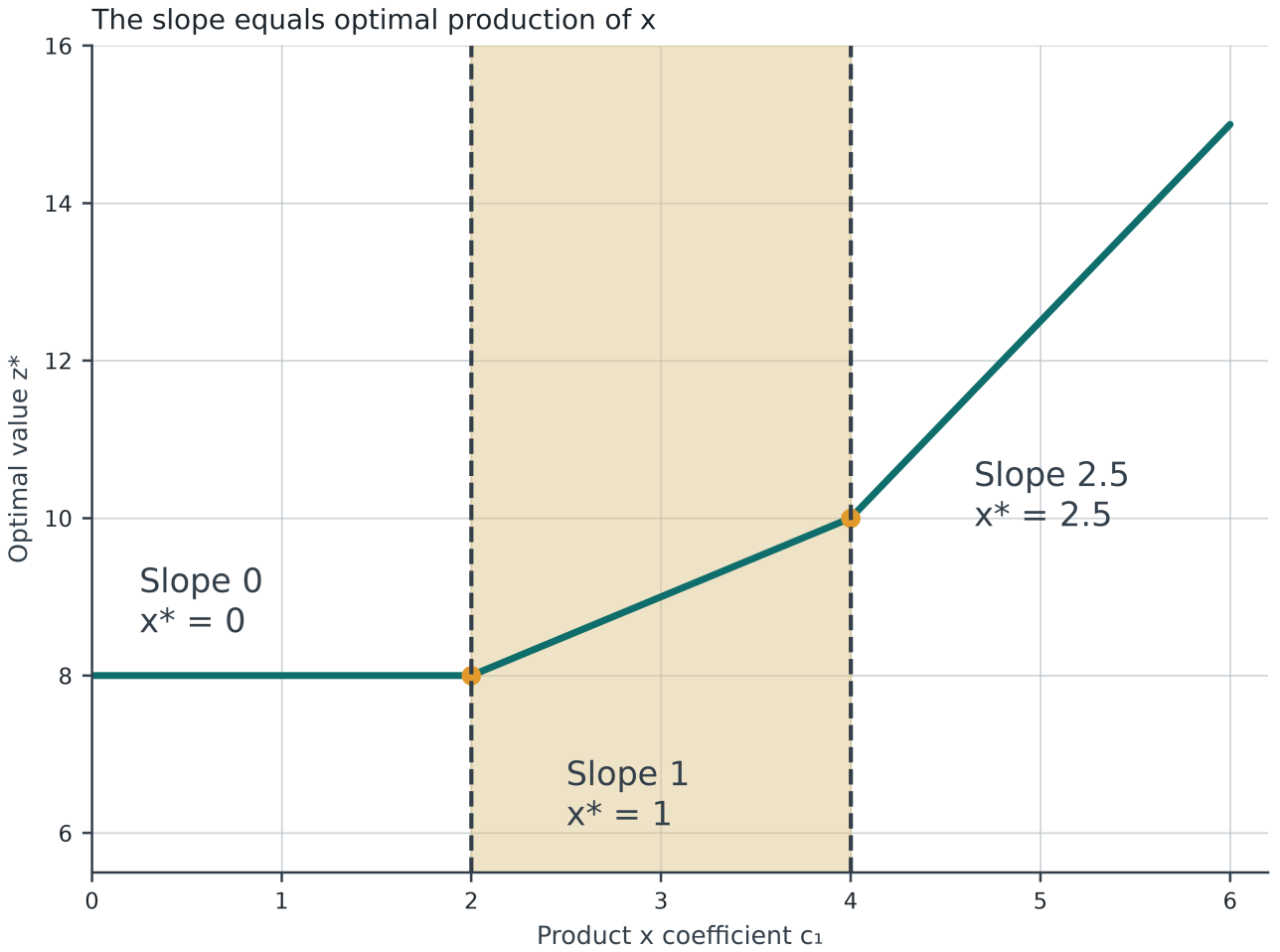
$c_1 \le 2$: $z^* = 8$, with $x^* = 0$.

$2 \le c_1 \le 4$: $z^* = c_1 + 6$.

$c_1 \ge 4$: $z^* = 2.5c_1$, with $x^* = 2.5$.

On each open interval, $dz^*/dc_1 = x^*$.

At 2 and 4, optimal edges give different one-sided slopes.



1. Return to the original production model

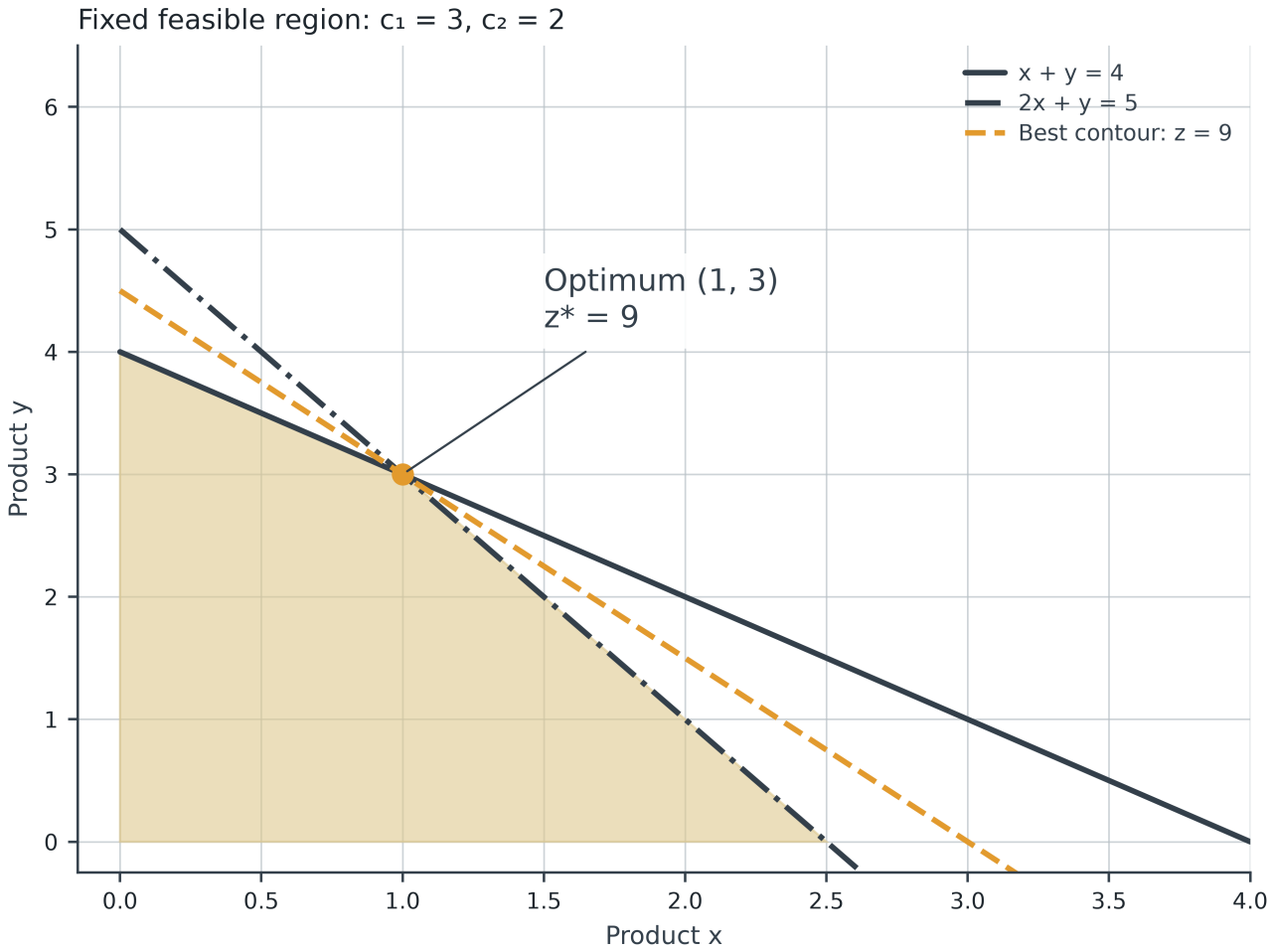
Maximize $z = 3x + 2y$.

$x + y \le 4$; $2x + y \le 5$; $x,y \ge 0$.

Baseline: $(x,y) = (1,3)$, $z^* = 9$.

Resource prices: $u = 1$, $v = 1$.

Could a new product improve this plan?



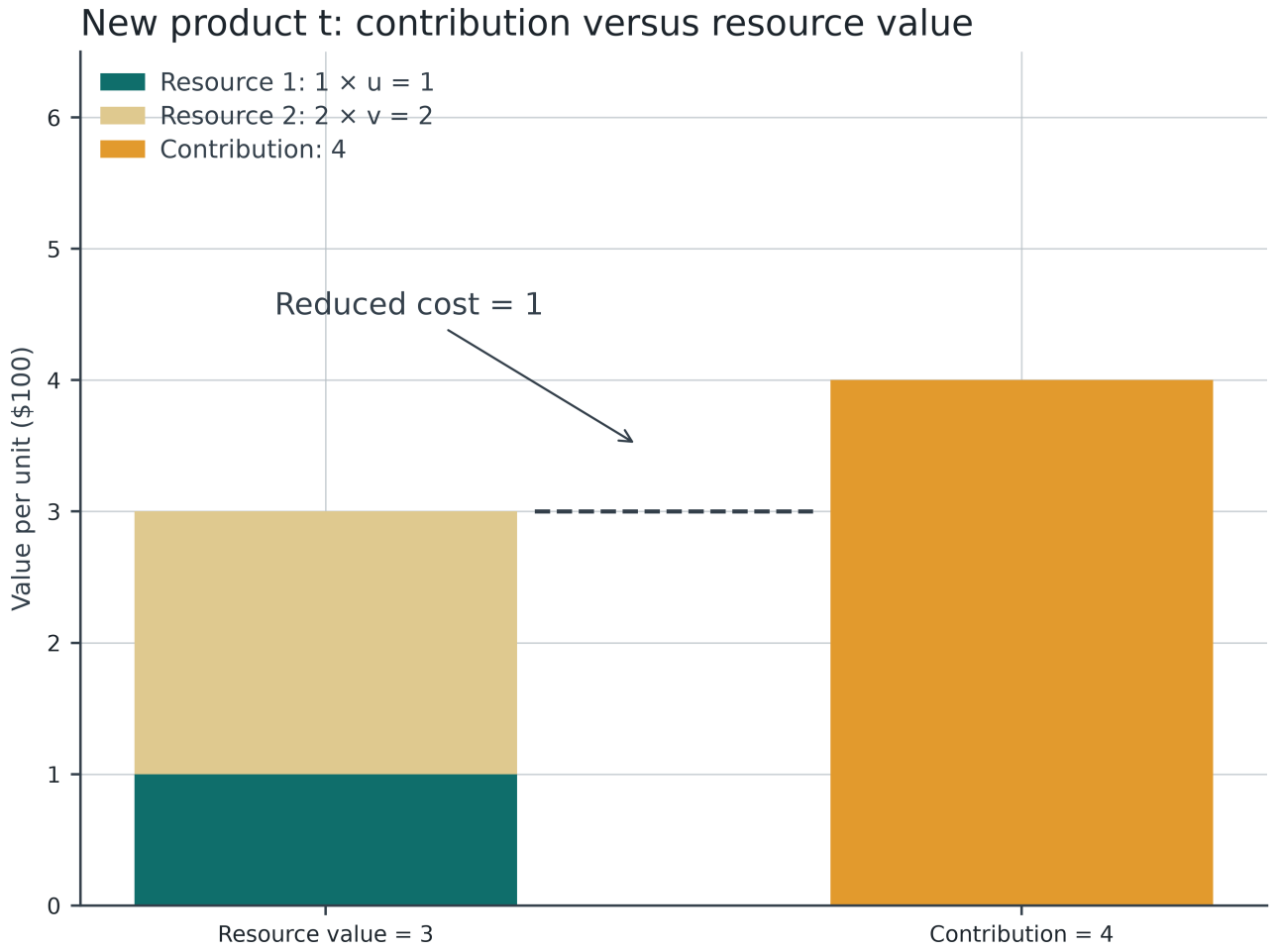
2. Price the resources used by the new product

New product t: contribution $c_t = 4$.

Each unit uses resource 1: 1; resource 2: 2.

At the old prices, resource value = 1u + 2v = 3.

Contribution 4 exceeds resource value 3 by 1.



3. Derive reduced cost from a feasible exchange

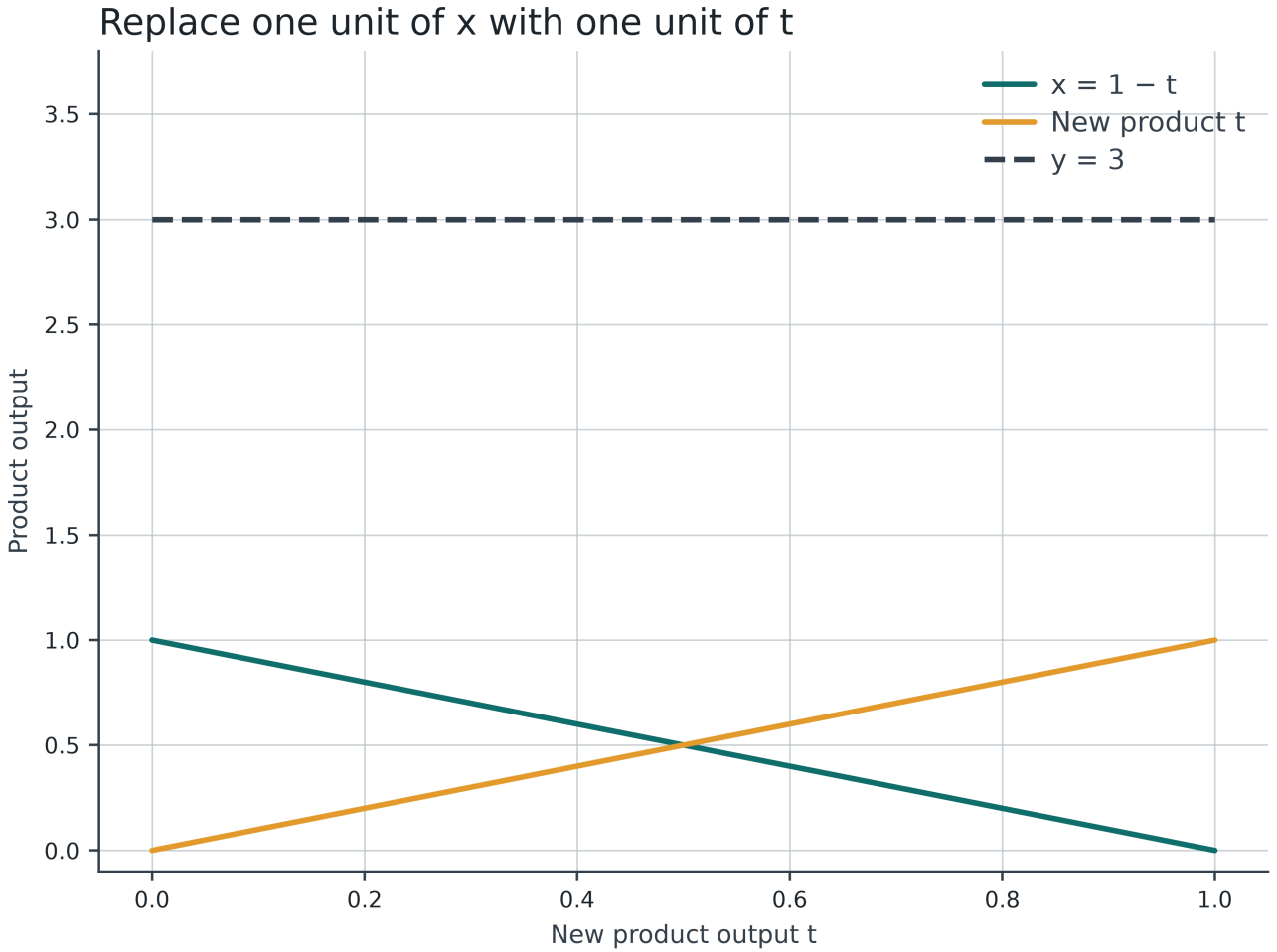
Keep both resources fully used:

$x + y + t = 4$; $2x + y + 2t = 5$.

Subtract: $x + t = 1$, so $x = 1 - t$ and $y = 3$.

$z = 3(1-t) + 2(3) + 4t = 9 + t$.

Thus $\bar{c}_t = c_t - (u+2v) = 4 - 3 = 1$.



4. The feasible step ends when x reaches zero

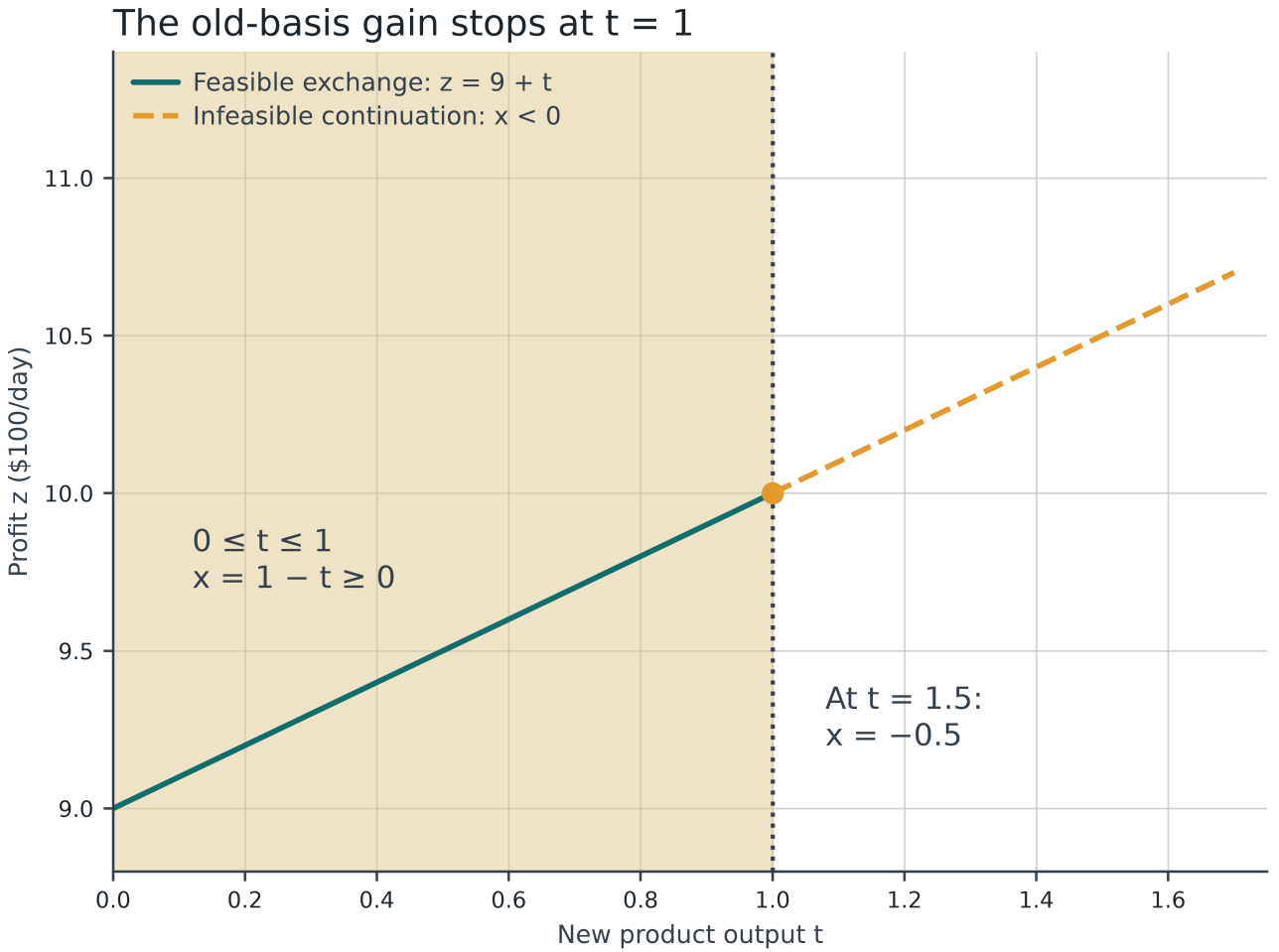
$x = 1 - t \geq 0$ and $t \geq 0$ require $0 \leq t \leq 1$.

At $t = 1$: $(x,y,t) = (0,3,1)$, $z = 10$.

At $t = 1.5$, the same rule gives $x = -0.5$.

So $z = 9 + t$ cannot be extrapolated past $t = 1$.

The basis must change when x leaves.



5. The new product invalidates the old certificate

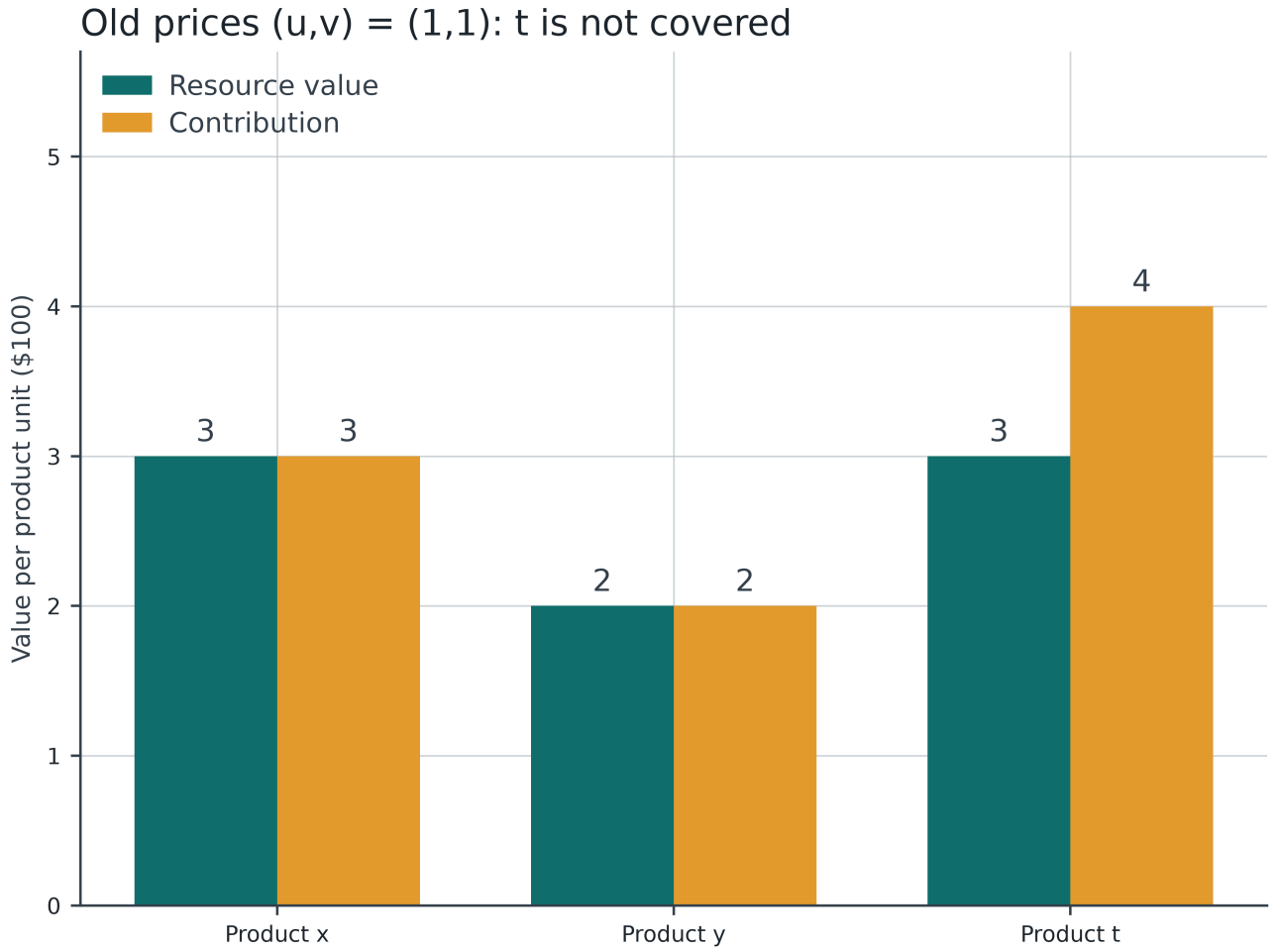
Max 3x + 2y + 4t; x,y,t ≥ 0.

x+y+t ≤ 4; 2x+y+2t ≤ 5.

Dual: u,v ≥ 0; u+2v ≥ 3; u+v ≥ 2; u+2v ≥ 4.

Old prices (1,1) fail: 3 < 4 for t.

So the old upper bound 9 is invalid.



6. Reoptimize and certify the new maximum

Choose new dual prices (u,v) = (0,2).

Resource values: x → 4 ≥ 3; y → 2; t → 4.

Every feasible plan has z ≤ 4u + 5v = 10.

Plan (0,3,1) attains 10, so it is optimal.

New reduced costs (x,y,t) = (−1,0,0).



7. More than one new production plan is optimal

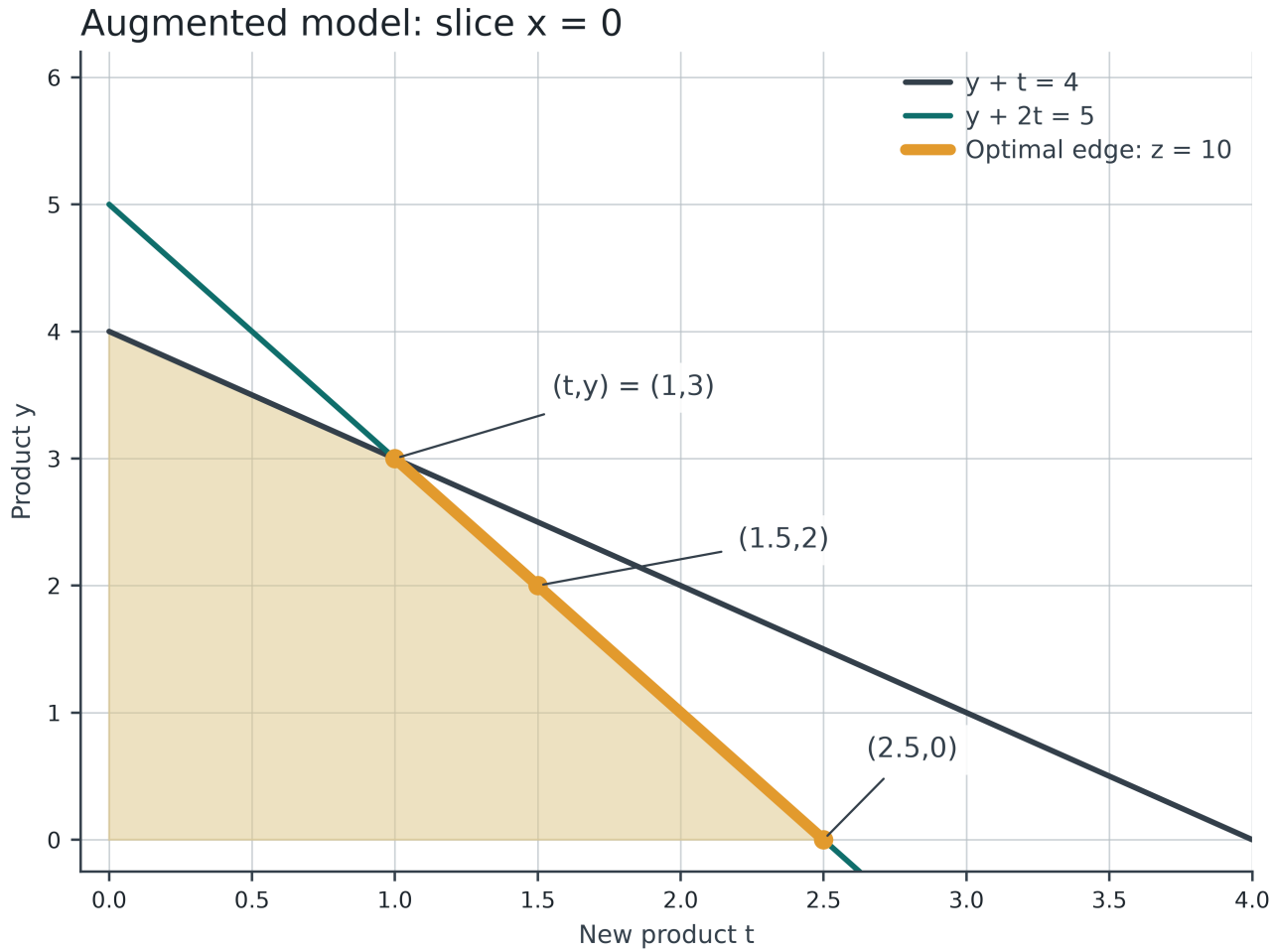
With x = 0: y+t ≤ 4 and y+2t ≤ 5.

On y+2t = 5, profit 2y+4t = 10.

Feasibility gives 1 ≤ t ≤ 2.5 and y = 5−2t.

Examples: (0,3,1), (0,2,1.5), (0,0,2.5).

Beyond t=1, reallocate y; profit stays at 10.



8. Compare three possible product contributions

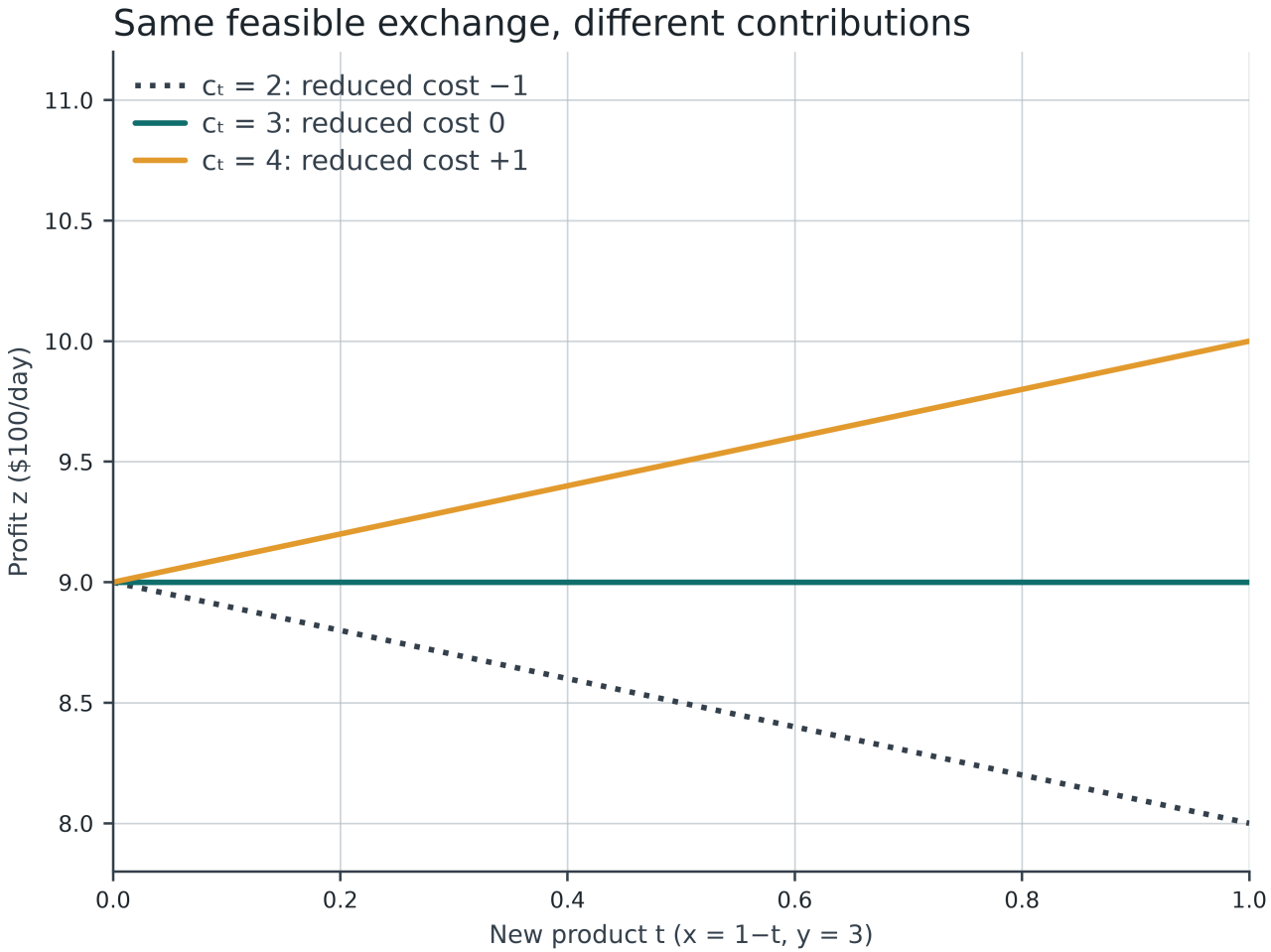
Keep the new column fixed at (1,2).

At old prices: $\bar{c}_t = c_t - 3$.

$c_t = 2$: $\bar{c}_t = -1$; optimal t = 0, z* = 9.

$c_t = 3$: $\bar{c}_t = 0$; feasible exchange keeps z* = 9.

$c_t = 4$: $\bar{c}_t = +1$; reoptimize to z* = 10.



9. A reduced-cost decision checklist

For max: $\bar{c}_j = c_j - a_j^T \pi$, where $\pi = (u,v)$.

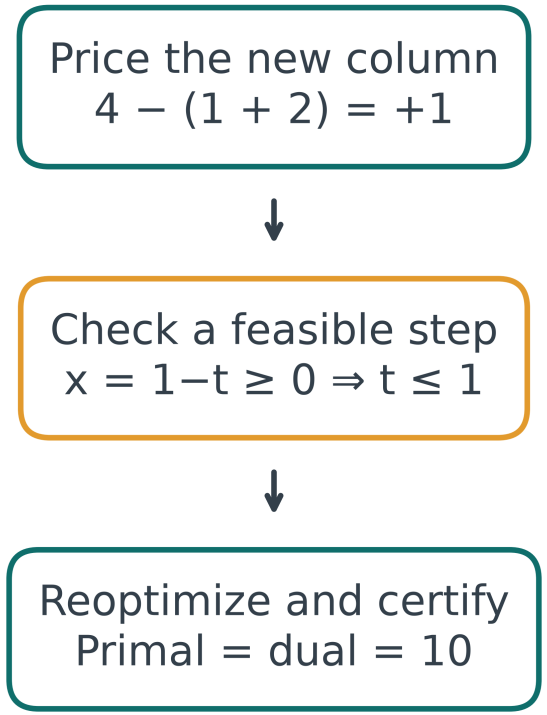
1. Price the column at the current dual prices.

2. Check a feasible step and its limit.

3. Reoptimize; check a new dual certificate.

Check solver conventions: min reduced costs use the opposite optimality sign.

From screening to a verified new optimum



Read solver output with the sign convention

Output	Expected value	Meaning
success	True	Optimal termination
x	[1, 3]	Production
fun	−9	Minimized negative contribution
ineqlin.residual	[0, 0]	Unused capacity
ineqlin.marginals	[−1, −1]	Sensitivity of the minimized objective

Return to the original two-product LP. SciPy minimizes $-3x-2y$; max-profit shadow prices = $-$marginals = (1,1).

Check status, feasibility, objective, and a small RHS change.

Practice: capacity purchase decision

Original LP, but an extra 0.5 unit/day of resource 1 costs $30/day.

A. Predict production and gross contribution after purchase.

B. Calculate net benefit and decide whether to buy.

C. Would the same shadow price justify adding 2 units/day?

Capacity purchase: worked solution

A. $b_1 = 4.5 \rightarrow (x,y) = (0.5,4)$, $z^* = 9.5 = \$950$/day.

B. Gross gain = \$50/day; cost = \$30/day; net gain = \$20/day → buy.

C. $b_1 = 6 \rightarrow (x,y) = (0,5)$, $z^* = 10 = \$1{,}000$/day.

Gross gain = \$100/day. Buy only if the extra cost is below \$100/day.

Exit ticket: explain, calculate, certify

1. Why can an LP have an optimal edge?

2. For the original LP, test dual prices $(u,v)=(1,0)$.

3. Give a feasible primal–dual pair with equal value.

4. Predict z^* at $b_1=4.2$; explain why extrapolating to $b_1=6$ fails.

Exit ticket: answers and reasoning

1. A contour parallel to a feasible edge can be optimal along the whole edge.

2. (1,0) fails $1+2\cdot0\geq3$ and $1+0\geq2$; $w=4$ is not a valid bound.

3. Primal (1,3), dual (1,1): $z=w=9$.

4. $z^*(4.2)=9.2$; at $b_1=6$ the basis changes and $z^*=10$.

What an LP solution should tell us

A feasible production plan and its objective value.

A dual bound that meets that value.

Complementarity that explains active products and resources.

Sensitivity ranges that limit any resource-value claim.

References and optional follow-up

Boyd and Vandenberghe, Convex Optimization: Sections 4.3, 5.2, 5.5, and 5.6.

SciPy documentation: linprog(method="highs"), marginals and residuals.

Optional Python practice: reproduce these examples with scipy.optimize.linprog.